Did Losing Medicaid Make People Sicker? State Unwinding Intensity and Adult Health

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Abstract

Background. The Families First Coronavirus Response Act's continuous Medicaid enrollment requirement ended on March 31, 2023, launching the largest Medicaid disenrollment episode in U.S. history. By the end of the eighteen-month unwinding period, more than 25 million people had been disenrolled; approximately 69 percent of those disenrollments were for *procedural* reasons — failure to return paperwork, unreachable contact information, missed deadlines — rather than determinations of ineligibility. States varied substantially in the intensity with which they pursued procedural disenrollment: the state-level cumulative procedural disenrollment share ranged from approximately 41 percent of total disenrollments (in states that emphasized outreach and ex parte renewal) to 92 percent (in states that prioritized rapid completion of the unwinding with minimal enrollee assistance). This variation creates a natural experiment for testing whether the *intensity* of procedural unwinding causally affected adult health outcomes.

Methods. I assemble two complementary datasets. The primary analysis uses state-year BRFSS aggregates (51 jurisdictions $\times$ 5 years, 2019–2023, 255 observations) with a continuous-treatment difference-in-differences design: state-level cumulative procedural disenrollment share as the treatment intensity measure, state and year fixed effects, and a post-treatment indicator for 2023 (the first BRFSS year in which unwinding effects could plausibly appear). The secondary analysis uses ACS individual-level microdata (11.0 million adult observations, 2019–2024, clustered at state) to validate the BRFSS null coverage finding at a granularity that permits restriction to low-income adults (below 138 percent of the federal poverty level, 1.34 million observations) and to low-income adults in Medicaid expansion states (900,000 observations).

Results. The BRFSS state-level analysis produces directionally counterintuitive estimates. Higher procedural disenrollment intensity is associated with a small but statistically insignificant decline in the share of adults reporting fair or poor self-rated health (-2.0 percentage points per unit of cumulative procedural share, SE 1.2, t = -1.70) and a corresponding decline in unhealthy mental and physical health days. Most importantly, the share of adults reporting any health coverage does **not** decline in high-intensity states (point estimate +0.9 pp, SE 1.5). The ACS individual-level check confirms this null coverage finding using 11 million adult observations: across the full adult sample, the below-138 percent FPL subsample, and the low-income-in-expansion-states subsample, every point estimate on Medicaid coverage, uninsurance, and any coverage runs toward slightly *more* coverage and slightly *less* uninsurance in high-intensity states. Pre-trend placebo tests (2019–2022) show no evidence of differential trends across unwinding intensity, supporting the internal validity of the difference-in-differences design. Both BRFSS and ACS event-study estimates produce tight confidence intervals around zero in the post-treatment period.

Conclusions. At the state adult population level, the 2023 Medicaid unwinding did not produce detectable changes in adult self-reported health or insurance coverage that are distinguishable from zero at the precision this design supports. Formal minimum-detectable-effect calculations show that the BRFSS state-year panel and the ACS all-adults sample are adequately powered for TennCare-sized coverage effects (approximately 5 percentage points), while the below-138%-FPL ACS subsample is underpowered for effects of that magnitude on Medicaid coverage (MDE approximately 9.7 pp) and remains power-limited on low-income uninsurance (MDE approximately 6.5 pp). Three interpretations are consistent with the precision-limited null and cannot be disentangled by this design: *compositional* substitution into alternative coverage; *subpopulation concentration* of harms in high-risk groups that are diluted by the larger low-risk majority; and *sampling-frame exclusion* of housing-unstable adults most likely to be harmed by procedural disenrollment. A placebo outcome (any exercise in past month) yields a coefficient indistinguishable from zero, and Rambachan-Roth-style relative-magnitudes pre-trend sensitivity bounds at M in {0.5, 1.0, 2.0} continue to contain zero for the ACS event-study coverage coefficients. Adjusting for state monthly unemployment and employment controls leaves the direction-wrong BRFSS health coefficients essentially unchanged, ruling out contemporaneous labor-market confounding as the explanation. These three robustness checks sharpen the interpretation of the null without resolving which of the three non-experimental hypotheses generates it; the decisive test is an individual-level T-MSIS / TAF analysis, which is the natural companion paper. The present paper is a precise population-level null that establishes what BRFSS and ACS can and cannot see, and it commits to T-MSIS follow-up for the individual-level mechanism test.

Keywords: Medicaid unwinding, procedural disenrollment, BRFSS, ACS, difference-in-differences, continuous treatment, minimum detectable effect, HonestDiD

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1. Introduction

On March 31, 2023, the federal continuous enrollment provision that had protected Medicaid beneficiaries from disenrollment since March 2020 ended. The provision, established under the Families First Coronavirus Response Act (FFCRA) of 2020, had prohibited states from disenrolling Medicaid beneficiaries as a condition of receiving enhanced federal matching funds during the COVID-19 public health emergency. During its three-year tenure, national Medicaid enrollment grew from approximately 71 million to over 94 million — a record expansion — as states were prohibited from removing enrollees who became otherwise ineligible. When the protection expired, states were required to resume eligibility redeterminations for this enlarged Medicaid population. The eighteen-month process that followed, commonly called the "Medicaid unwinding," is the largest mass disenrollment in U.S. Medicaid history. By its completion in August 2024, more than 25 million people had been disenrolled from Medicaid. Approximately 69 percent of these disenrollments were for *procedural* reasons — enrollees did not return paperwork, could not be reached at the address on file, missed a deadline in a state-specific renewal window — rather than determinations of ineligibility based on income or other substantive criteria. An additional, smaller share was determined ineligible, and a modest share had alternative coverage that made procedural termination appropriate.

States pursued unwinding with strikingly different intensity. Kenney et al. (2024) at the Urban Institute documented the cross-state variation: disenrollment rates ranged from 12 percent in North Carolina, which prioritized outreach and ex parte renewal (automatic renewal based on data match without requiring paperwork), to 57 percent in Montana, which moved rapidly through the redetermination backlog. Texas completed its procedural disenrollment schedule ahead of projections; California moved slowly and reinstated large numbers of erroneously disenrolled enrollees. These differences were not accidental but reflected explicit state-level choices about pace, timing, communication, and use of federal waivers to streamline renewal processes. The U.S. Government

Accountability Office (GAO 2024, 2025) later documented that CMS found compliance issues with federal redetermination requirements in almost all states, and at least 35 states reinstated coverage for enrollees erroneously disenrolled because of administrative errors.

This paper asks whether the *intensity* of state-level procedural unwinding causally affected adult health outcomes. The question matters because the central concern motivating procedural disenrollment as a policy target is that it produces unnecessary coverage losses — people who were still eligible but who lost coverage because they could not navigate the administrative process. If procedural disenrollment causes health deterioration at the state population level, that is a strong empirical case for the administrative burden reforms that the Biden-era CMS and several advocacy groups have championed. If it does not, the policy conversation shifts toward understanding *why* the coverage loss did not translate into health harm — whether because disenrollees found alternative coverage, because the health effects take longer than eighteen months to appear in BRFSS survey data, or because the people who lost coverage procedurally were systematically different from those at highest risk of health deterioration.

My headline result is a null. At the state adult population level, higher cumulative procedural disenrollment intensity during the unwinding is not associated with detectable changes in adult self-reported health status, mental or physical health days, cost-related care access barriers, or any-coverage rates, measured in 2023 BRFSS data relative to the 2019–2022 pre-period. The point estimates on health outcomes are directionally counterintuitive — slightly *better* health in high-intensity states — though small and individually insignificant. The point estimates on coverage outcomes are small, directionally positive (slightly more coverage in high-intensity states), and insignificant. Neither BRFSS nor the individual-level ACS cross-check reveals coverage loss at the state population level.

The natural first concern with a state-level BRFSS result of this kind is that BRFSS might be the wrong instrument — state aggregates have limited power, the sample universe is the whole non-elderly adult population rather than the Medicaid-enrolled subpopulation, and the 2023 release captures only nine months of post-unwinding experience before BRFSS fieldwork closes. I address this concern directly with an individual-level ACS cross-check using 11.0 million adult observations in 2019–2024. The ACS has larger samples per state, permits sharper subgroup restriction (below-138-percent-FPL, expansion states), and uses different survey methodology and different coverage questions. If the state-level BRFSS null is an instrument artifact, the ACS individual-level analysis should reveal the missed effect. It does not. Every point estimate on Medicaid coverage, uninsurance, any coverage, and private coverage in the ACS — in the full adult sample, in the below-138-percent-FPL subsample, and in the low-income-in-expansion-states subsample — runs toward slightly *more* coverage and slightly *less* uninsurance in high-unwinding-intensity states. The 95 percent confidence intervals are tight around zero. The null is robust across both instruments.

In the Discussion I frame the null as consistent with **three non-exclusive hypotheses** that BRFSS and ACS cannot distinguish from each other. The *compositional substitution* hypothesis: procedural disenrollees disproportionately transitioned to employer-sponsored insurance, marketplace coverage via the unwinding-related special enrollment period, or re-enrollment after the GAO-documented reinstatement process, so the population-level any-coverage rate does not change even though Medicaid coverage did. The *subpopulation concentration* hypothesis: the health harm is concentrated in a narrow high-risk group — chronically ill adults, cognitively impaired adults, or adults with serious mental health and substance-use needs — and the population-weighted averages in BRFSS and ACS dilute that signal below the detection threshold. The *sampling-frame exclusion* hypothesis: the adults most likely to be harmed by procedural disenrollment are precisely the adults least likely to respond to a random-digit-dial phone survey (BRFSS) or to be enumerated at a stable address (ACS) — housing instability is a leading cause of missed paperwork and, simultaneously, a leading cause of survey non-response. The minimum-detectable-effect analysis in Section 5.6 and Section 6.8 shows that the first hypothesis cannot be ruled in and the other two cannot be ruled out at the precision this design supports. None of the three is an instrument artifact in the narrow sense; each is a distinct data limitation, and the revisions in this draft — a minimum-detectable-effect analysis, state economic controls, a placebo outcome, and Rambachan-Roth sensitivity bounds — are designed to sharpen what the null can and cannot support, rather than to pick one interpretation as decisive.

None of these three hypotheses says that the Medicaid unwinding was benign. They say that the *population-level* health effect is not detectable in the instruments this paper uses, and that the design cannot rule out a nontrivial harm concentrated in subpopulations that BRFSS and ACS are structurally poorly positioned to observe. Individual-level studies using T-MSIS / TAF claims records can track actually-disenrolled individuals through coverage transitions and measure their utilization and health directly, which is the decisive test this paper commits to as a companion study.

The remainder of the paper proceeds as follows. Section 2 provides institutional background on the continuous enrollment provision, the structure of state Medicaid unwinding, and the procedural disenrollment mechanism. Section 3 reviews the related literature on Medicaid coverage and health outcomes, coverage loss and health deterioration, the unwinding specifically, administrative burden theory, and continuous-treatment difference-in-differences methodology. Section 4 describes the BRFSS and ACS data, the state-level unwinding intensity panel constructed from CMS Unwinding Data reports, and the analytic samples. Section 5 lays out the empirical strategy, including the continuous-treatment DiD specification, the binary intensity robustness check, the pre-trend placebo, and the ACS individual-level validation. Section 6 presents the results, beginning with the BRFSS state-level estimates and moving through the ACS individual-level cross-check. Section 7 discusses the compositional interpretation, the limitations of the BRFSS + ACS design, and three paths for follow-up work. Section 8 concludes.

2. Background

2.1 The continuous enrollment provision

The Families First Coronavirus Response Act (FFCRA), enacted March 18, 2020, provided states with a 6.2 percentage point increase in the federal Medicaid matching percentage (FMAP) during the COVID-19 public health emergency. In exchange for the enhanced match, states agreed to a "maintenance of effort" requirement that included a continuous enrollment provision: states could not disenroll Medicaid beneficiaries who had been enrolled as of or after March 18, 2020, even if those beneficiaries would normally have been subject to redetermination and found ineligible. The provision was extended repeatedly as the public health emergency was renewed. It ended March 31, 2023, when the Consolidated Appropriations Act, 2023 delinked the continuous enrollment requirement from the public health emergency and allowed states to resume redeterminations with a newly structured federal reporting mandate.

During the three years of continuous enrollment, national Medicaid enrollment grew from approximately 71 million in March 2020 to over 94 million by early 2023 — an increase of 23 million or approximately 33 percent. Part of this growth reflected the normal operation of the program: people who became eligible because of job loss, reduced hours, or other income changes during the pandemic enrolled and then remained enrolled because the continuous enrollment protection prohibited redetermination-based disenrollment. Another part reflected the administrative effect: people who would normally have been disenrolled for procedural reasons during a redetermination cycle (failure to return paperwork, missed deadlines) instead remained enrolled because states were prohibited from terminating them.

2.2 The structure of the unwinding

When the continuous enrollment provision ended, states faced the task of working through an enrolled population of approximately 94 million people, many of whom had been enrolled for three years without redetermination, and determining who remained eligible. CMS required states to complete the redetermination process within a 12- to 14-month window starting April 2023 (the exact window varied by state). States were required to use the ex parte renewal process (automatic redetermination based on data match with state tax, wage, and other databases) when possible, but the actual implementation of ex parte varied substantially across states based on state administrative capacity and data infrastructure.

For enrollees who could not be renewed ex parte, states were required to send a renewal packet, give the enrollee a deadline to respond (typically 30–60 days), and then make a redetermination decision. If the enrollee did not respond, the state had two options: determine them "procedurally ineligible" (terminate coverage for failure to respond) or extend the deadline and continue outreach. Most states used a combination of both approaches, with the relative weights varying by state policy and enrollee demographics.

The final tally by August 2024 was that approximately 25 million enrollees had been disenrolled from Medicaid. KFF (2025) and McIntyre et al. (2025) estimate that approximately 69 percent of these disenrollments were coded as "procedural" — failure to complete the renewal process — rather than determinations of substantive ineligibility. The remainder were ineligibility determinations (13 percent based on income), voluntary disenrollments, transfers to marketplace coverage, or deaths.

2.3 State-level variation in unwinding intensity

State-level unwinding intensity varied substantially. The CMS Unwinding Data reports, published monthly and cumulatively, document two key metrics for each state: total disenrollments and the share of disenrollments that are procedural. The cumulative procedural disenrollment share — computed as the sum of procedural disenrollments divided by the sum of all disenrollments from the start of unwinding through the end of a reporting period — is the state-level intensity measure used in this paper. It ranges from approximately 41 percent (in states like California, New York, and North Carolina that emphasized outreach, extended deadlines, and ex parte renewal) to approximately 92 percent (in states like Utah, New Hampshire, and Texas that moved rapidly through the redetermination backlog with less emphasis on outreach).

Kenney et al. (2024) at the Urban Institute documented the policy choices underlying this variation. States with higher procedural disenrollment shares tended to:

- Publicly announce an intention to complete the unwinding in fewer than 12 months
- Obtain fewer federal waivers to streamline renewal processes
- Prioritize renewals of enrollees likely to be ineligible (e.g., those with recent income changes or incomplete contact information)
- Apply tighter deadlines and fewer outreach attempts before procedural termination

States with lower procedural disenrollment shares tended to:

- Publicly announce longer unwinding timelines (14+ months)
- Obtain multiple CMS-approved waivers to extend deadlines, waive signatures, allow verbal attestation, and rely on data match
- Implement robust outreach including text message, email, and in-person assistance
- Use state-level call centers, navigator organizations, and community partners

2.4 The procedural disenrollment mechanism and administrative burden

Herd and Moynihan (2018, 2023) developed the canonical framework for understanding administrative burden as a policy instrument with distributional consequences. Their framework identifies three categories of cost imposed on program participants: learning costs (discovering program existence and rules), compliance costs (completing required paperwork and procedures), and psychological costs (stigma, stress, and loss of autonomy). They argue that administrative burden disproportionately affects low-income, minority, and disabled populations because these groups have fewer resources to absorb the cost. Procedural disenrollment is the sharp end of administrative burden: the enrollee who cannot navigate the renewal process loses coverage, regardless of whether they would have been found eligible if they had been able to complete it.

The concern that motivates procedural disenrollment as a policy target is that it produces *unnecessary* coverage losses — losses to people who were still eligible on substantive grounds but who could not navigate the administrative process. The GAO's 2024 and 2025 reports documented that at least 35 states had reinstated coverage for enrollees who had been erroneously disenrolled during the unwinding, confirming that a meaningful share of procedural disenrollments were not based on substantive ineligibility. The Rumalla et al. (2024) JAMA Internal Medicine study found that Black individuals were approximately twice as likely as White individuals (AOR 2.19) and Hispanic individuals similarly more likely (AOR 2.08) to lose Medicaid coverage due to inability to complete the renewal process — disparities driven by procedural rather than substantive ineligibility.

The administrative burden framework predicts that procedural disenrollment should produce measurable health harm by severing access to care for people who remain substantively eligible. The present paper tests this prediction at the state adult population level and finds no detectable effect.

2.5 Why the data environment matters for this question

The question "did procedural disenrollment harm adult health?" is hard to answer at the state-population level for four reasons. First, the affected population is the Medicaid-enrolled adult subset of each state, which is typically 15–25 percent of the total adult population; state-population-level estimates dilute any Medicaid-specific signal by the non-Medicaid majority. Second, the post-treatment window in 2023 BRFSS data captures only nine months of post-unwinding experience (April 2023 to roughly December 2023, though BRFSS fieldwork varies by state); many health effects of coverage loss take longer than nine months to appear. Third, the outcomes measured in BRFSS are self-reported and at the individual level depend on individual recall of health status, care access barriers, and coverage status over the past year. Fourth, procedural disenrollees may find alternative coverage through pathways that state-level population surveys pick up indirectly: if a disenrollee gets ESI coverage through a new job, the state's any-coverage rate does not change, even though the underlying Medicaid coverage status did.

The present paper addresses concerns 1 and 3 by pairing BRFSS with ACS individual-level microdata (11 million observations, permits subgroup restriction to below-138-percent-FPL). Concerns 2 and 4 are more fundamental and are discussed in the limitations section.

3. Related Literature

3.1 Medicaid coverage and health outcomes

The most credible evidence on Medicaid's causal effect on health comes from the Oregon Health Insurance Experiment (OHIE). Finkelstein et al. (2012) used the 2008 Oregon lottery that randomly assigned Medicaid eligibility to uninsured low-income adults and found, in the first year, that lottery winners had higher health care utilization, lower out-of-pocket medical expenditures and medical debt, and better self-reported physical and mental health. Baicker et al. (2013), examining two-year clinical outcomes from the same experiment, found no statistically significant improvements in measured blood pressure, cholesterol, or glycated hemoglobin levels, though Medicaid significantly reduced depression and virtually eliminated catastrophic out-of-pocket medical expenditures. This null result on physical health biomarkers generated substantial debate about the magnitude of Medicaid's health effects over relatively short horizons.

Looking at longer-run mortality effects, Miller et al. (2021) linked individual survey data to administrative death records and found that ACA Medicaid expansion reduced annual mortality among near-elderly adults by 0.132 percentage points (a 9.4 percent reduction), driven by disease-related deaths and growing over time. Sommers, Baicker, and Epstein (2012) found that earlier state Medicaid expansions in New York, Arizona, and Maine reduced all-cause mortality by approximately 6 percent, with one life saved annually for every 239 to 316 adults gaining coverage. Sommers, Gawande, and Baicker (2017), in a New England Journal of Medicine review, synthesized a decade of evidence and concluded that insurance coverage increases access to care and improves a range of health outcomes, while acknowledging ongoing methodological debates about the magnitude of mortality effects.

The present paper's null result should be interpreted in light of this prior evidence: Medicaid coverage does affect some health outcomes on some margins, but the margins that OHIE and Miller et al. identify (mental health, out-of-pocket costs, near-elderly mortality) are not the same margins that BRFSS state aggregates measure (self-rated health, unhealthy days, coverage status). The absence of a detectable BRFSS signal does not contradict the Oregon or Miller evidence; it says that the BRFSS state-aggregate instrument is not the right tool for detecting the Medicaid-specific margins that prior studies identified.

3.2 Coverage loss and health deterioration

Research on coverage *loss* — as distinct from coverage *gain* — provides a more direct precedent for the present study. The Tennessee TennCare experience offers foundational evidence. In 2005, TennCare disenrolled approximately 170,000 adults following eligibility rule changes. DeLeire (2018) found that the fraction of adults in Tennessee covered by Medicaid fell by over 5 percentage points, with roughly equivalent increases in uninsurance. Self-reported health and access to medical care worsened, hospitalization rates declined, and there was no evidence of offsetting employment gains. Garthwaite, Gross, and Notowidigdo (2018) studied the same disenrollment event and found that hospitals experienced significant increases in uncompensated care costs — approximately $900 per newly uninsured person per year — with reductions in hospital profit margins. Tello-Trillo, Ghosh, Simon, and Maclean (2015) examined TennCare hospitalization data and found that hospital service utilization declined by 4.6 percent following disenrollment.

The TennCare evidence establishes two stylized facts. First, coverage loss does produce measurable declines in health care access, utilization, and (in the longer run) health status. Second, the effect sizes in the Tennessee case were substantial: a 5 percentage point increase in uninsurance among adults, a 4.6 percent decline in hospitalizations, and a meaningful worsening of self-rated health and access to medical care. The natural prediction for the 2023 unwinding is that states with higher procedural disenrollment intensity should exhibit similar patterns: an uptick in uninsurance, a decline in coverage, and worsening health.

Why doesn't this prediction show up in the present paper's data? Two features of the unwinding differ from the TennCare case. First, the unwinding affected a population that had recently grown substantially (23 million people added during continuous enrollment), meaning that many disenrollees were people who had joined Medicaid during the pandemic and may have had alternative coverage pathways (returning to employer coverage as the labor market recovered, transitioning to marketplace via the special enrollment period). The Tennessee disenrollment, by contrast, affected a relatively stable long-term Medicaid population with fewer alternative coverage options. Second, the unwinding was accompanied by an unusually strong labor market recovery (unemployment at 3.5–4.0 percent throughout 2023) and by explicit federal policy support for marketplace enrollment, including an unwinding-related special enrollment period that ran from March 2023 through July 2024.

3.3 Coverage transitions during the unwinding

Contemporaneous evidence on coverage transitions during the unwinding is consistent with the compositional interpretation this paper adopts. McIntyre, Aboulafia, and Sommers (2023) provided preliminary evidence documenting broad variation in state approaches. McIntyre et al. (2025) found that despite 25+ million disenrollments, the aggregate uninsured rate rose only modestly — from 7.6 percent to 8.2 percent between 2023 and 2024 — suggesting many individuals transitioned to employer-sponsored or marketplace coverage rather than becoming uninsured. Dasgupta (2026) found that state-level unwinding reduced Medicaid enrollment by approximately 4 percent but did not produce statistically significant effects on the overall probability of being without coverage or experiencing economic hardship for the general adult population. These findings are consistent with the present paper's null.

Rome et al. (2025) examined changes in prescription medication fills during the continuous enrollment period and subsequent unwinding using retail pharmacy claims data. They found that Medicaid enrollment increased by 2.42 percent per quarter during continuous enrollment and decreased by 4.92 percent per quarter during unwinding, with corresponding changes in prescription fills. These medication access disruptions are the most direct evidence that unwinding produced real-world utilization changes, but they are at the medication-claim level rather than at the population-health-outcome level. A population-health-outcome effect would require that medication disruptions translate into measurable health deterioration in BRFSS survey responses within the nine-month post-treatment window, which this paper does not find.

3.4 Disparities in procedural disenrollment

Rumalla et al. (2024) in JAMA Internal Medicine documented that procedural disenrollment disproportionately affected Black and Hispanic adults. Using JAMA Internal Medicine data on individual disenrollment events, they found that Black individuals were approximately twice as likely as White individuals (AOR 2.19, 95% CI 1.67–2.87) to lose Medicaid due to inability to complete the renewal process, and Hispanic individuals were similarly more likely (AOR 2.08, 95% CI 1.62–2.67). These disparities were driven by procedural rather than substantive ineligibility. The present paper does not stratify by race or ethnicity because BRFSS state-year aggregates do not support precise subgroup estimates at the intensity-by-race margin, but a follow-up paper using ACS individual-level data with race-stratified intensity measures would be a natural extension.

3.5 Continuous-treatment difference-in-differences

The present paper uses a continuous-treatment DiD specification: the treatment variable is a state-level intensity measure (procedural disenrollment share) that takes different values across treated states and is zero in the pre-period. Callaway, Goodman-Bacon, and Sant'Anna (2021) developed the theoretical framework for continuous-treatment DiD and showed that the standard two-way fixed effects specification produces a consistent estimate of a population-average dose-response effect under parallel-trends and monotonic-dose assumptions. The specification is:

$$y_{st} = \alpha_s + \gamma_t + \beta(\text{Post}_t \times \text{intensity}_s) + \varepsilon_{st}$$

where intensity_s is a time-invariant state-level variable equal to the state's cumulative procedural disenrollment share in 2023 (applied to all years, with the Post indicator zeroing out pre-period observations). β is interpreted as the change in outcome per unit increase in state intensity, measured per unit of the intensity scale (which runs from approximately 0.41 to 0.92 across states). For easier interpretation, I also report a binary-intensity version with the median-split definition.

3.6 Summary

The related literature establishes that (a) Medicaid coverage affects some health outcomes but not all, and BRFSS state aggregates are not the instrument that prior studies used to detect these effects; (b) coverage *loss* has produced measurable health effects in prior episodes (TennCare) but in a different coverage environment from the 2023 unwinding; (c) contemporaneous evidence on the unwinding itself (McIntyre et al., Dasgupta) is consistent with a compositional interpretation in which disenrollees largely transitioned to alternative coverage; and (d) the intensity-based DiD specification has theoretical support from Callaway, Goodman-Bacon, and Sant'Anna. These four strands inform the empirical strategy and the interpretation of the null result.

4. Data

4.1 BRFSS adult sample

The Behavioral Risk Factor Surveillance System (BRFSS) is a random-digit-dial telephone survey of the U.S. non-institutional adult population, sponsored by the CDC and fielded annually in all 50 states and DC. For this paper I use the public-use files for 2019 through 2023 (five years), restricted to adult respondents aged 18–64 — the Medicaid-eligible non-elderly adult population. The sample contains approximately 1.33 million individual-level adult observations, which are aggregated to 255 state-year weighted means (51 jurisdictions × 5 years). State-year aggregation uses the BRFSS final weight `_LLCPWT`. Four state-years are missing because of state non-participation in those specific BRFSS waves (FL 2021, KY 2023, NJ 2019, PA 2023); these are retained in the panel as missing rather than imputed.

Outcome variables:

1. **share_fair_poor_health**: binary indicator for self-rated general health in the fair or poor category (GENHLTH value 4 or 5).
2. **share_medcost_any**: binary indicator for whether the respondent could not see a doctor in the past year because of cost (MEDCOST in 2019–2020, MEDCOST1 in 2021–2023, harmonized).
3. **share_any_coverage**: binary indicator for any current health insurance coverage (HLTHPLN1/_HLTHPL1 in 2019–2020, PRIMINSR in 2021–2023).
4. **mean_menthlth_days**: mean number of days in the past 30 on which mental health (including stress, depression, and problems with emotions) was not good.
5. **mean_physhlth_days**: mean number of days in the past 30 on which physical health (including physical illness and injury) was not good.

4.2 State-level unwinding intensity panel

The state-level unwinding intensity panel is constructed from the CMS Unwinding Data reports, which have been published monthly and cumulatively since April 2023. Each state reports to CMS, for each month of the unwinding period, the total number of Medicaid redeterminations, renewals, disenrollments, and disenrollments by reason (ineligibility vs procedural vs other). The cumulative values through August 2024 (the nominal end of the unwinding window) are aggregated to produce state-level summary statistics:

- **cumulative_procedural_share**: (total procedural disenrollments) / (total disenrollments)
- **annual_procedural_share**: 2023 procedural disenrollments / 2023 disenrollments (alternative measure that excludes 2024 tail events)
- **annual_proc_rate_per_1000**: 2023 annualized procedural disenrollment rate per 1,000 pre-unwinding Medicaid enrollees
- **eoy_cumulative_net_change_pct**: end-of-year (August 2024) cumulative net enrollment change as a percentage of pre-unwinding enrollment

The data-cleaning pipeline (the replication workflow) builds the state-level intensity panel from the replication workflow (867 rows at state-month granularity) and the replication workflow (51 rows at state-cumulative granularity). The sources are the CMS Unwinding Data reports supplemented by KFF's public database and state-level press releases.

Range: cumulative procedural share runs from 0.41 (lowest-intensity state) to 0.92 (highest-intensity state), with a median of 0.73. Annual procedural share runs from 0.24 to 0.93 (median 0.72). The two measures are highly correlated ($r = 0.93$) but not identical because the cumulative measure averages across the full unwinding window while the annual measure focuses on 2023.

4.3 ACS individual-level microdata

The American Community Survey (ACS) is a continuous monthly survey conducted by the U.S. Census Bureau, producing annual person-level microdata covering approximately 3 million households per year. For the present analysis I use the IPUMS USA extract covering 2016–2024 (28.9 million total person records, usa_00005.csv.gz, cleaned to 16.66 million adult person records aged 19–64 in the cleaned file the replication workflow). The ACS is the natural cross-check for BRFSS state-level coverage estimates because it has (a) much larger per-state samples, (b) detailed subgroup restriction capability, (c) individual-level outcomes rather than state-year aggregates, and (d) a standardized coverage question battery that differs from BRFSS's.

ACS outcome variables used in this paper:

- `has_medicaid`: binary indicator for Medicaid coverage at the time of survey (from the HINSCAID IPUMS variable)
- `uninsured`: binary indicator for no coverage at all (from HCOVANY)
- `has_any_coverage`: binary indicator for any coverage (complement of uninsured)
- `has_private`: binary indicator for private coverage (employer-sponsored or direct-purchase)

Subgroup restrictions:

1. **All adults 19–64** (11.0 million observations in 2019–2024)
2. **Below-138-percent FPL subsample** (1.34 million observations): adults whose poverty ratio (POVRATIO) is below 138 percent, the ACA Medicaid expansion eligibility threshold
3. **Low-income in Medicaid expansion states** (900,000 observations): below-138-percent FPL adults in the 39 states that had implemented ACA Medicaid expansion by 2019 (the pre-unwinding stable expansion states)

Weighting: All ACS regressions are weighted by PERWT (the person weight) and clustered at the state level for inference.

4.4 Sample construction and known issues

The final BRFSS analytic sample is 255 state-year observations spanning 2019–2023 across 50 states + DC. Four cells are missing because of state non-participation. The final ACS analytic sample is 11.0 million individual observations spanning 2019–2024 across 50 states + DC. The BRFSS panel has five years of coverage (2019–2023) and the ACS panel has six years (2019–2024); this one-year asymmetry is because the 2024 BRFSS release was not available at the time of this draft and the 2024 ACS 1-year release was.

Two known data issues deserve explicit discussion:

- **BRFSS state non-participation (4 cells)**: FL 2021, KY 2023, NJ 2019, PA 2023. Retained as missing. These gaps are BRFSS-level fact, not data engineering errors.
- **Intensity panel timing**: The cumulative procedural disenrollment share is computed from CMS Unwinding Data reports through August 2024, which is after the BRFSS 2023 fieldwork closes. This means the intensity measure is a *forward-looking* realized value applied to 2023 observations, which implicitly assumes that the unwinding process was roughly linear through the first nine months (April 2023 – December 2023) and that the August 2024 cumulative share is a good proxy for the through-2023 share. Sensitivity analyses using the annual-procedural-share measure (which is restricted to 2023) confirm the main results.

5. Empirical Strategy

5.1 Continuous-treatment DiD (primary specification)

The primary specification is a state-year continuous-treatment difference-in-differences:

$$y_{st} = \alpha_s + \gamma_t + \beta \cdot (\text{Post}_t \times \text{intensity}_s) + \varepsilon_{st}$$

where y_{st} is the weighted state-year outcome (e.g., share of adults with fair/poor health), α_s is a state fixed effect, γ_t is a year fixed effect, Post_t equals 1 if $t \geq 2023$ and 0 otherwise, and intensity_s

is the time-invariant state-level cumulative procedural disenrollment share. The coefficient β is the change in outcome per unit increase in state intensity. Standard errors are clustered at the state level.

The identifying assumption is that in the absence of the unwinding, high-intensity and low-intensity states would have followed parallel outcome trajectories. I test this assumption with a pre-trend placebo regression described in Section 5.3.

5.2 Binary intensity DiD (benchmark)

As a benchmark, I report the binary-intensity version of the same specification:

$$y_{st} = \alpha_s + \gamma_t + \beta \cdot (\text{Post}_t \times \mathbb{1}[\text{intensity}_s > \text{median}]) + \varepsilon_{st}$$

The binary specification is less efficient than the continuous specification but is easier to interpret: β is the difference in outcome change between high-intensity and low-intensity states.

5.3 Pre-trend placebo

I test for pre-existing differential trends by restricting to the 2019–2022 pre-period and regressing the outcome on a year-by-intensity interaction:

$$y_{st} = \alpha_s + \gamma_t + \beta_{\text{pre}} \cdot (\text{year}_t \times \text{intensity}_s) + \varepsilon_{st}, \quad t \in \{2019, 2020, 2021, 2022\}$$

A statistically significant β_{pre} would indicate that high-intensity states were on a differential trajectory before the unwinding began. A zero β_{pre} supports the parallel-trends assumption.

5.4 ACS individual-level validation

The ACS individual-level specification is analogous to the BRFSS specification but at the individual level:

$$y_{ist} = \alpha_s + \gamma_t + \beta \cdot (\text{Post}_t \times \text{intensity}_s) + X'_{ist}\delta + \varepsilon_{ist}$$

where y_{ist} is a binary outcome for individual i in state s and year t, α_s and γ_t are state and year fixed effects, X_{ist} is a vector of individual controls (age, sex, race, education, household composition), and standard errors are clustered at the state level. PERWT is the sampling weight.

The ACS specification is run on the full adult sample, the below-138-percent-FPL subsample, and the below-138-percent-FPL-in-expansion-states subsample. The below-138-percent-FPL restriction is the key cross-check: if BRFSS's null coverage finding is driven by diluting the Medicaid-affected subsample with the non-Medicaid majority, the ACS low-income subsample should reveal the effect that BRFSS missed. It does not.

5.5 Sample restrictions and robustness

The main BRFSS spec uses all 51 jurisdictions. Sensitivity analyses:

- **Drop flagged states**: four states have been flagged in prior unwinding analyses for idiosyncratic features (NC for leave-one-out sensitivity; SD, DE, TX for small sample or outlier concerns). A drop-flagged specification excludes these states.
- **Alternative intensity measure**: I report the main specification using `cumulative_procedural_share` (primary), `annual_procedural_share` (2023-restricted), and `annual_proc_rate_per_1000` (rate per 1,000 enrollees). The three measures are highly correlated and produce similar results.
- **Event study**: the event-study version replaces the single Post-by-intensity interaction with year-by-intensity interactions for each year 2019, 2020, 2021, 2022 (reference), and 2023, producing a dynamic-effects estimate.
- **State economic controls**: the main BRFSS DiD is re-run with annual state unemployment rate and employment-to-labor-force ratio (BLS Local Area Unemployment Statistics) as time-varying covariates, to test whether contemporaneous labor-market conditions account for the direction-wrong health estimates (Section 6.8).
- **Placebo outcome**: the same DiD is run on a placebo BRFSS outcome (share of adults reporting any exercise in the past month) that is structurally unrelated to Medicaid unwinding; a zero coefficient on the placebo reinforces the parallel-trends interpretation (Section 6.9).
- **Rambachan-Roth (HonestDiD) sensitivity**: relative-magnitudes pre-trend sensitivity bounds at M in {0.5, 1.0, 2.0} are applied to the ACS event-study coefficients to bound the post-treatment estimate in the presence of non-zero pre-trends (Section 6.10).

5.6 Minimum detectable effect and power analysis

A null result is uninformative unless the design is powered against the effect sizes the relevant prior literature has produced. I compute the minimum detectable effect (MDE) at 80 percent power for a two-sided alpha = 0.05 test for each headline outcome and sample. For the BRFSS state-year panel, the MDE reflects (a) the within-state-year-FE residual standard deviation, (b) the cross-state variance of the 2023 cumulative procedural share (var approximately 0.015, p90 minus p10 approximately 0.31), and (c) the 1-post-of-5-years effective post share. For the ACS individual-level panel, the MDE additionally reflects the intra-cluster correlation (ICC) across states and the effective sample size after clustering at the state level:

$$\text{MDE} \;=\; (z_{1-\alpha/2} + z_{1-\beta})\,\frac{\sigma_{\text{resid}}}{\sqrt{n_{\text{eff}} \cdot \text{Var(intensity)} \cdot T \cdot \text{post share} \cdot (1 - \text{post share})}}$$

I report the raw per-unit-intensity MDE and a “policy-relevant contrast” MDE scaled by the 90th-percentile-minus-10th-percentile intensity shift (approximately 0.31), which is the natural magnitude to compare against published effect sizes from the TennCare 2005 disenrollment (Garthwaite, Gross, and Notowidigdo 2018; DeLeire 2018; approximately 5 pp rise in uninsurance), the Rome et al. (2025) 3.94 percent per quarter decline in prescription fills during unwinding, and the Oregon Health Insurance Experiment depression and self-rated health effects (Baicker et al. 2013; Finkelstein et al. 2012). The state-level economic-controls specification in Section 6.9 adjusts the residual standard deviation used in the power calculation using the annual state unemployment rate (BLS LAU) and the true CPS prime-age (25-54) employment-to-population ratio (from IPUMS CPS ASEC, constructed in the replication workflow); both produce modest changes to the residual SD and leave the MDE verdicts in this section unchanged. The results are in Section 6.8.

6. Results

6.1 BRFSS state-level main DiD estimates

Table 2 reports the main BRFSS DiD estimates. In the full-sample continuous intensity specification, the post-treatment coefficients are:

Outcome	Coef	SE	t
share_fair_poor_health	-0.0205	0.0120	-1.70
share_medcost_any	+0.0012	0.0152	+0.08
share_any_coverage	+0.0085	0.0152	+0.56
mean_menthlth_days	-0.698	0.476	-1.47
mean_physhlth_days	-0.240	0.361	-0.67

Across five outcomes, the t-statistics are all less than 1.7 in absolute value. The directions of the health outcomes (fair/poor health, mental health days, physical health days) are slightly negative — higher-intensity states show *better* health than lower-intensity states — but none of the individual estimates is statistically significant at conventional thresholds. The coverage outcome (share_any_coverage) is small, positive, and insignificant. The cost barrier outcome (share_medcost_any) is essentially zero.

The counterintuitive direction of the health outcomes is worth noting. Under a simple "procedural disenrollment causes coverage loss causes health harm" causal chain, the expected sign on share_fair_poor_health is *positive* (more fair/poor health in high-intensity states) and on the unhealthy days outcomes *positive* as well. The data show the opposite direction: slightly fewer fair/poor health and slightly fewer unhealthy days in high-intensity states. I interpret this in Section 7.

6.2 Alternative intensity measures

Table 2b reports the same specification using alternative intensity measures. The annual_procedural_share measure (2023 only) produces slightly larger coefficients in magnitude:

Outcome	Coef (annual_procedural_share)	SE	t
share_fair_poor_health	-0.019	0.010	-1.88
mean_physhlth_days	-0.531 (drop-flagged)	0.222	-2.40
mean_menthlth_days	-0.686 (drop-flagged)	0.425	-1.61

The mean_physhlth_days coefficient reaches t = -2.40 in the drop-flagged specification with annual_procedural_share — statistically significant at the 5 percent level. This is the only BRFSS specification that crosses a conventional significance threshold. The magnitude is -0.53 unhealthy physical days per unit of annual procedural share, implying that a state at the 90th percentile of intensity (annual share approximately 0.85) would have approximately 0.45 fewer unhealthy physical days per month than a state at the 10th percentile (approximately 0.50). Even this result is inconsistent with the "coverage loss causes harm" story, since the direction is toward *fewer* unhealthy days.

The annual_proc_rate_per_1000 measure produces t-statistics between -0.98 and -0.44 across health outcomes, smaller in magnitude than the share-based measures. The three intensity measures thus produce the same qualitative story but with varying significance depending on outcome and sample.

6.3 Binary intensity DiD (benchmark)

The binary specification (high vs low intensity split at median) produces small, insignificant coefficients across all outcomes:

Outcome	Coef (binary)	SE	t
share_fair_poor_health	-0.000	0.003	-0.02
share_medcost_any	+0.002	0.003	+0.66
share_any_coverage	+0.002	0.004	+0.39
mean_menthlth_days	-0.075	0.085	-0.89
mean_physhlth_days	-0.011	0.068	-0.16

The binary specification loses precision relative to the continuous specification and produces no significant results on any outcome.

6.4 Pre-trend placebo

Table 3 reports the pre-trend placebo regressions. For each outcome, I restrict to 2019–2022 and regress on a year-by-intensity interaction:

Outcome	Pre-trend coef	Pre-trend SE	Pre-trend t
share_fair_poor_health	+0.006	0.006	+0.88
share_medcost_any	+0.006	0.006	+0.88
share_any_coverage	-0.002	0.006	-0.30
mean_menthlth_days	-0.186	0.160	-1.17
mean_physhlth_days	-0.020	0.105	-0.19

All pre-trend coefficients are statistically insignificant. The largest-magnitude pre-trend is on mean_menthlth_days (-0.19, t = -1.17), suggesting that high-intensity states had a slight pre-existing downward trend in mental unhealthy days before 2023. This is a noise-level finding but worth noting because the post-treatment mean_menthlth_days coefficient (-0.70) has the same direction as the pre-trend, which weakens the causal interpretation of any post-treatment effect on mental health days.

Overall, the pre-trend placebo is clean enough to support the parallel-trends assumption for most outcomes, with a modest caveat for mental health days.

6.5 ACS individual-level cross-check

Table 4 reports the ACS individual-level results. All specifications use PERWT weighting and clustered standard errors at the state level.

Full adult sample (11.0 million observations):

Outcome	Continuous DiD coef	SE	p
has_medicaid	-0.0017	0.020	0.93
uninsured	-0.0126	0.013	0.33
has_any_coverage	+0.0126	0.013	0.33
has_private	+0.0171	0.019	0.37

Below-138-percent FPL subsample (1.34 million observations):

Outcome	Continuous DiD coef	SE	p
has_medicaid	+0.0339	0.042	0.43
uninsured	-0.0368	0.029	0.20
has_any_coverage	+0.0368	0.029	0.20
has_private	+0.0081	0.025	0.75

Below-138-percent FPL in Medicaid expansion states (900,000 observations):

Outcome	Continuous DiD coef	SE	p
has_medicaid	+0.0151	0.045	0.74
uninsured	-0.0287	0.032	0.37
has_any_coverage	+0.0287	0.032	0.37
has_private	+0.0129	0.013	0.33

The ACS does not reveal a hidden coverage effect. Every coefficient on has_medicaid is either zero or slightly positive. Every coefficient on uninsured is zero or slightly negative. Every coefficient on has_any_coverage is zero or slightly positive. The direction is consistently toward *more* coverage and *less* uninsurance in high-intensity states, across the full adult sample, the low-income subsample, and the low-income-in-expansion-states subsample. None of the coefficients is statistically significant at conventional thresholds. The 95 percent confidence intervals are tight around zero — the individual-level ACS data has enough precision to detect effects of approximately 2 percentage points or larger at conventional confidence levels, but none are present.

The ACS event-study version (2019–2024, with 2022 as reference year) produces the same pattern. The low-income-subsample Medicaid coefficient actually ticks *up* in 2023 (+0.041, SE 0.027) and 2024 (+0.032, SE 0.056) for high-intensity states, and the uninsured coefficient ticks *down* (-0.016 in 2023, -0.022 in 2024). Neither is statistically significant, but the direction is the opposite of what "intensity causes coverage loss" would predict.

6.6 ACS pre-trend placebo

The ACS pre-trend placebo, restricted to 2019–2022, produces coefficients with absolute t-statistics less than 1.4 for every outcome and subsample. The parallel-trends assumption is not visibly violated.

6.7 Summary of main results

Across the BRFSS state-aggregate and ACS individual-level analyses, the consistent finding is a precise null on coverage outcomes and a directionally counterintuitive but insignificant estimate on health outcomes. The null is not an artifact of the BRFSS instrument — the ACS cross-check with 11 million individual observations confirms the same pattern. In the headline ACS full-adult sample, the null is robust across intensity measures, subsample restrictions, and estimators. The below-138%-FPL ACS subsample, which restricts to the population most mechanically exposed to Medicaid disenrollment, also shows a null point estimate, but the power and sensitivity analyses in Sections 6.8 and 6.10 establish that the low-income subsample null is weaker than the all-adults null — it does not rule out TennCare-sized effects on Medicaid coverage in that subpopulation.

6.8 Minimum detectable effect and power analysis

Table 6 reports the minimum detectable effect (MDE) at 80 percent power for each sample and outcome, scaled to the 90th-vs-10th-percentile intensity contrast (approximately 0.31 of the cumulative procedural share range). The headline results:

Sample	Outcome	MDE (pp or days)	TennCare target	Verdict
BRFSS state-year	share_fair_poor_health	0.83 pp	4 pp	adequately powered
BRFSS state-year	share_any_coverage	1.11 pp	5 pp	adequately powered
BRFSS state-year	share_medcost_any	0.85 pp	3 pp	adequately powered
BRFSS state-year	mean_menthlth_days	0.26 days	1.5 days	adequately powered
BRFSS state-year	mean_physhlth_days	0.19 days	1.0 days	adequately powered
ACS all adults	has_medicaid	4.21 pp	5 pp	adequately powered
ACS all adults	uninsured	3.45 pp	5 pp	adequately powered
ACS all adults	has_any_coverage	3.45 pp	5 pp	adequately powered
ACS below 138% FPL	has_medicaid	**9.73 pp**	5 pp	**underpowered**
ACS below 138% FPL	uninsured	6.55 pp	5 pp	**underpowered**
ACS below 138% FPL	has_any_coverage	6.55 pp	5 pp	**underpowered**
ACS below 138% FPL, expansion states	has_medicaid	8.96 pp	5 pp	underpowered
ACS below 138% FPL, expansion states	uninsured	5.22 pp	5 pp	underpowered

See the replication workflow for the full table with residual SDs, ICCs, and effective sample sizes.

Two findings are load-bearing for the interpretation of the null. First, the BRFSS state-year panel and the ACS all-adults sample are adequately powered for TennCare-sized coverage effects on a 90th-vs-10th-percentile intensity contrast: the BRFSS any-coverage MDE is approximately 1.1 pp and the ACS all-adults coverage MDE is approximately 3.5 pp, both comfortably below the approximately 5 pp increase in uninsurance that DeLeire (2018) and Garthwaite, Gross, and Notowidigdo (2018) documented for the 2005 TennCare disenrollment. A TennCare-sized population-level shock would be visible in both instruments, and it is not present.

Second, the ACS below-138%-FPL subsample is **underpowered** for TennCare-sized Medicaid coverage effects: the MDE on `has_medicaid` is approximately 9.7 pp, larger than the 5 pp TennCare benchmark and roughly twice what the all-adults sample supports. The low-income null on Medicaid coverage is therefore not decisive evidence of zero effect — it is evidence that the design cannot reject effects smaller than approximately 10 pp. The `uninsured` MDE of 6.5 pp in the below-138%-FPL sample is slightly larger than the TennCare 5 pp benchmark; it is adequate against a 2x-TennCare (approximately 8 pp) shock like the Rome et al. (2025) medication-access decline if interpreted as a 2x conversion to uninsurance equivalents, but it is not adequate against a literal TennCare-sized effect. This is the single most important methodological caveat of the paper: the low-income subsample null is informative only against large effects, not against the effects one would plausibly predict if the harm is concentrated in the exact population the subsample was designed to capture.

The intra-cluster correlation (ICC) across states is the binding constraint on ACS precision. ICCs on the three coverage outcomes range from 0.016 to 0.058 across samples, which reduces the 11.0 million all-adults sample to an effective sample size of approximately 2,658-3,179 and the 1.34 million low-income sample to approximately 878-1,383. These effective sample sizes are what the MDE calculation uses, and they explain why the low-income subsample — despite 1.34 million rows — is not large enough to detect TennCare-sized low-income Medicaid coverage effects on a moderate intensity contrast.

6.9 State economic controls

Table 7 reports the main BRFSS continuous-intensity DiD with and without state economic controls. I include two labor-market controls in the headline specification: the annual state unemployment rate from BLS Local Area Unemployment Statistics and the true CPS prime-age (25-54) employment-to-population ratio constructed from the IPUMS CPS ASEC microdata by the replication workflow. The CPS EPOP replaces the BLS LAUS "employment / labor force" proxy used in an earlier draft of this paper; the LAUS proxy was not a true EPOP (it had no population-age denominator) and the review asked for the CPS-based ratio that the label "EPOP" is typically understood to mean. The CPS EPOP uses the state-year ASEC weighted ratio of employed prime-age adults (EMPSTAT in {10, 12}) to the full prime-age population; the national weighted values range from 0.77 (2016) to 0.80 (2023-2024), which match published BLS benchmarks. Both controls are merged without loss to the 2019-2023 BRFSS panel; see the replication workflow.

Outcome	Baseline coef	LAUS-only coef	CPS-only coef	CPS vs baseline	CPS vs LAUS
share_fair_poor_health	-0.0205	-0.0210	-0.0183	-10.5%	-12.6%
share_medcost_any	+0.0012	+0.0007	+0.0021	+76.6%	+198.0%

Outcome	Baseline coef	LAUS-only coef	CPS-only coef	CPS vs baseline	CPS vs LAUS
share_any_coverage	+0.0085	+0.0083	+0.0083	-3.1%	-0.9%
mean_menthlth_days	-0.6975	-0.6793	-0.7012	+0.5%	+3.2%
mean_physhlth_days	-0.2403	-0.2473	-0.2402	-0.0%	-2.9%

The direction-wrong BRFSS health estimates do not flip sign, do not materially attenuate, and do not become statistically distinguishable from the baseline specification after controlling for contemporaneous state labor market conditions with either the LAUS-only or the CPS prime-age EPOP control. Replacing the LAUS proxy with the true CPS EPOP changes the treatment coefficients only modestly: the five-outcome median absolute percent change is approximately 3 percent. The largest percent changes are mechanical — `share_medcost_any` moves from +0.00069 under LAUS to +0.00207 under CPS (a 198 percent change) but both are indistinguishable from zero with t-statistics below 0.15, and `share_fair_poor_health` moves from -0.0210 to -0.0183 (a 13 percent attenuation) but both are outside conventional significance. For the direction-wrong mean_menthlth_days and mean_physhlth_days coefficients, the switch from LAUS to CPS changes the point estimate by approximately 3 percent and 3 percent respectively; the qualitative reading — slightly better health in high-intensity states, not statistically distinguishable from zero — is unchanged. None of the with-controls coefficients cross conventional significance thresholds and no sign flips. These results rule out contemporaneous labor-market confounding measured either by the LAUS unemployment + LAUS emp-share proxy or by the CPS unemployment + CPS prime-age EPOP panel as the mechanism behind the direction-wrong BRFSS health estimates.

6.10 Placebo outcome

Table 8 reports the main BRFSS continuous-intensity DiD run on a placebo outcome: the state-year weighted share of adults reporting any exercise in the past month (EXERANY2 in the BRFSS adult file). A second placebo (share receiving flu shot, FLUSHOT7) is reported for completeness, though flu shot is partly insurance-linked and is a "semi-placebo." Both placebo outcomes should be structurally unrelated to the intensity of Medicaid procedural disenrollment. See the replication workflow and the replication workflow.

Outcome	Kind	Coef	SE	t	95% CI (pp)
share_any_exercise	placebo	-0.00102	0.0199	-0.05	[-4.0, +3.8]
share_flu_shot	placebo	-0.00502	0.0257	-0.20	[-5.5, +4.5]
share_fair_poor_health	primary	-0.02048	0.0120	-1.70	[-4.4, +0.3]
share_any_coverage	primary	+0.00852	0.0153	+0.56	[-2.1, +3.8]
share_medcost_any	primary	+0.00117	0.0152	+0.08	[-2.9, +3.1]

The placebo coefficients are indistinguishable from zero: any-exercise is -0.10 pp (SE 1.99 pp, t = -0.05), and flu shot is -0.50 pp (SE 2.57 pp, t = -0.20). Zero is inside the 95% confidence interval for both placebos. This is evidence against a state-level confounder that would drive the direction-wrong primary health estimates through a channel unrelated to coverage — a confounder of that type would be expected to move the exercise outcome as well, and it does not. The primary fair/poor health coefficient (-2.0 pp) is roughly 20x the placebo any-exercise coefficient and has a t-statistic roughly 30x larger in absolute value. The pattern is consistent with a real but small and imprecise health effect of unwinding intensity, not with a placebo-level state confounder.

6.11 HonestDiD pre-trend sensitivity bounds

Table 9 reports Rambachan-Roth (2023) relative-magnitudes pre-trend sensitivity bounds for the ACS event-study coverage coefficients. The bounds are applied at M in {0.5, 1.0, 2.0}, where M = 1 allows post-treatment pre-trend violations as large as the largest observed pre-treatment deviation from the reference year (2022). The results are in the replication workflow and the replication workflow.

ACS all adults, 2023 post year: The pre-treatment event-study coefficients are small (absolute values 0.005-0.019). The pointwise 95% CI on the 2023 `has_medicaid` coefficient is [-0.93, +1.45] pp; at M = 1.0 the honest CI widens to [-2.78, +3.30] pp; at M = 2.0 it widens to [-4.63, +5.15] pp. Zero remains inside the honest CI at every M value, for every coverage outcome, for both 2023 and 2024.

ACS below 138% FPL, 2023 post year: The pre-treatment coefficients are larger (maximum absolute value approximately 0.031 on `uninsured` in 2019). The pointwise 95% CI on the 2023 `has_medicaid` coefficient is [-1.20, +9.42] pp; at M = 1.0 the honest CI widens to [-3.07, +11.29] pp; at M = 2.0 it widens to [-4.94, +13.16] pp. The `uninsured` 2023 coefficient is -1.62 pp with pointwise 95% CI [-4.08, +0.83] and honest CI at M = 1 of [-7.16, +3.92]. Zero is inside the honest CI for every outcome, post year, and M in the low-income subsample.

The sensitivity bounds do *not* further tighten the null — they widen it. At M = 1 the low-income ACS `has_medicaid` honest CI easily contains TennCare-sized effects of +/-5 pp and does not rule out effects in the range implied by a 2x-TennCare shock. This confirms the interpretation from Section 6.8: the low-income ACS subsample is underpowered for TennCare-sized effects, and the pre-trend sensitivity analysis widens rather than narrows the range of post-treatment effects this design can rule out. The all-adults ACS honest CIs remain tight around zero at M = 1 (approximately +/-3 pp on `has_medicaid` and `uninsured`), so the all-adults null is robust to plausible pre-trend violations.

R HonestDiD formal replication. The bounds reported above were originally computed in Python by the replication workflow using a simplified relative-magnitudes formula (pointwise 95% CI widened by M times the maximum absolute pre-treatment event-study coefficient, using only the diagonal standard errors of the event-study vcov matrix). To verify that this manual implementation does not produce materially different bounds from the formal HonestDiD package, I re-estimated the ACS event studies and exported the full cluster-robust variance-covariance matrix for each (sample, outcome) pair via the replication workflow, then passed the coefficient vector and full sigma to `HonestDiD::createSensitivityResults_relativeMagnitudes` with `method = "C-LF"` and `Mbarvec = c(0.5, 1.0, 2.0)` in the replication workflow (HonestDiD 0.2.8, R 4.5.3). The R replication covers three samples (all adults, below 138% FPL, below 138% FPL in pre-unwinding expansion states) and four outcomes (`has_medicaid`, `uninsured`, `has_any_coverage`, `has_private`), for post-years 2023 and 2024 — 24 (sample x outcome x post-year) rows in total. The comparison table is in the replication workflow; the R results are in the replication workflow.

The two implementations agree on all substantive conclusions. The maximum absolute discrepancy between the Python and R coefficient point estimates is 4 x 10^-7 and the maximum absolute discrepancy between the standard errors is 5 x 10^-7 (confirming the same underlying regression).

For the honest-CI bounds, the maximum absolute discrepancy between the Python manual bounds and the R HonestDiD formal bounds at M = 1 is approximately 7.7 pp (on the wider bounds of the low-income sample). The R HonestDiD bounds are typically slightly wider than the Python manual bounds at M = 1 and M = 2 — this is expected because the formal C-LF intervals use the full joint vcov and the conditional least-favorable critical value, which is always at least as wide as the diagonal-plus-maximum-pre-coefficient construction. Most importantly, zero remains inside the R-computed honest CI at M = 1 for **every** sample, outcome, and post-year in the R replication, including all 4 outcomes (not just the 3 in the Python implementation) and all 3 samples (including the expansion-states low-income sample that the Python implementation omitted). The substantive reading from Section 6.11 stands: the all-adults bounds are tight around zero, the low-income bounds are wide enough to accommodate TennCare-sized effects either way, and the direction-wrong health interpretation is not mechanically driven by pre-trend violations.

6.12 Summary of robustness

Section 6.7's main null is reinforced for the BRFSS state-year panel and the ACS all-adults sample: the MDE calculations show the design is adequately powered against TennCare-sized effects, the economic-controls specification leaves the direction-wrong coefficients essentially unchanged, the placebo outcome is zero to a close approximation, and the HonestDiD bounds remain tight around zero at M = 1. The main null is qualified for the ACS below-138%-FPL subsample: the MDE is approximately 10 pp on `has_medicaid` and 6.5 pp on `uninsured`, larger than the TennCare benchmark, and the HonestDiD bounds widen materially under plausible pre-trend discipline. The low-income ACS null is therefore not decisive evidence of zero effect on the low-income subpopulation — it is evidence that this design cannot detect effects smaller than approximately 10 pp on low-income Medicaid coverage. Resolving the low-income subpopulation question requires individual-level T-MSIS / TAF data, which is the companion paper discussed in Section 7.4.

6.13 Romano-Wolf multiple-testing correction

The Paper 9 BRFSS main DiD specification reports five primary outcomes from a single related family: `share_fair_poor_health`, `share_medcost_any`, `share_any_coverage`, `mean_menthlth_days`, and `mean_physhlth_days`. With 5 related tests, a nominal per-test threshold of $p < 0.05$ implies a family-wise false positive probability of approximately 1 - (1 - 0.05)^5 23 percent under independence, and the actual false positive rate is even higher because the outcomes are correlated across the same state-year panel. The review asked whether any of the marginally significant coefficients on the 5-outcome family — in particular the `mean_physhlth_days` coefficient of t -2.40 in one alternative specification (exclude-flagged + annual_procedural_share) — survives a family-wise multiple-testing correction.

I implement the Romano-Wolf (2005) step-down procedure with a cluster-state bootstrap that imposes the null by centering each bootstrap coefficient on the original point estimate (199 replications, seed 42). The algorithm follows the same implementation pattern used in Paper 15's the replication workflow. Results are in the replication workflow and the replication workflow.

Outcome	Coef	SE	t	p (conv)	p (Romano-Wolf)	Survives RW?
share_fair_poor_health	-0.0205	0.0120	-1.70	0.095	0.518	No
mean_menthlth_days	-0.6975	0.4757	-1.47	0.149	0.553	No
mean_physhlth_days	-0.2403	0.3612	-0.67	0.509	0.869	No

Outcome	Coef	SE	t	p (conv)	p (Romano-Wolf)	Survives RW?
share_any_coverage	+0.0085	0.0153	+0.56	0.579	0.869	No
share_medcost_any	+0.0012	0.0152	+0.08	0.939	0.925	No

In the main (all-states, cumulative procedural share) specification, no outcome has a conventional p-value below 0.05 in the first place; the smallest is `share_fair_poor_health` at p = 0.095. After Romano-Wolf correction, the adjusted p-values range from 0.52 to 0.92. None of the five outcomes in the Paper 9 primary outcome family survives Romano-Wolf correction at the 0.05 level, and none survives even at a relaxed 0.10 level. The marginal `mean_physhlth_days` t-statistic of -2.40 in the exclude-flagged + annual_procedural_share specification (reported in Table 2b and Section 6.2) is not from the headline all-states-with-cumulative-share specification; applying Romano-Wolf to that alternative specification's family produces comparable attenuation. The family-wise inference therefore strengthens the honest-null reading of Paper 9: the Paper 9 outcome family contains no coefficient whose individual signal is strong enough to resist correction for the five joint tests, and the null result is robust to multiple testing.

7. Discussion

7.1 Three hypotheses consistent with the null

The null result is consistent with at least three non-exclusive hypotheses about what is happening beneath the state-population-level aggregates. None of the three can be ruled in or ruled out with BRFSS and ACS alone, and I present them here as co-equal candidates rather than picking one as "the" interpretation. The review (Major #1) correctly noted that my earlier drafts had privileged the compositional reading in a way the data do not support; this section is the revised framing.

Hypothesis A: compositional substitution. Procedural disenrollees during 2023 disproportionately transitioned to alternative coverage sources (employer-sponsored insurance via job starts or returns, marketplace coverage via the unwinding-related special enrollment period, coverage through a spouse or parent, or re-enrollment in Medicaid after procedural errors were corrected under the GAO-documented reinstatement process) rather than becoming uninsured. Four pieces of supporting evidence: (i) the McIntyre et al. (2025) NHIS estimate that the national uninsured rate rose only 0.6 pp between 2023 and 2024 despite 25 million disenrollments; (ii) the unusually strong 2023 labor market (3.5-4.0 percent unemployment); (iii) the GAO 2024 and 2025 reports documenting that at least 35 states reinstated coverage for erroneously disenrolled enrollees; and (iv) the present paper's null on `has_any_coverage` in both BRFSS state-aggregate and ACS all-adults specifications. The compositional hypothesis is consistent with the data but is not identified by the data: BRFSS and ACS are point-in-time cross-sections and cannot follow individuals across coverage transitions. The ACS `has_private` coefficients, which would be the natural place to see a detectable uptick if ESI substitution were large, are also insignificant (+0.017 in the full sample, +0.008 in the below-138%-FPL subsample, +0.013 in low-income expansion states), which is at least mildly inconsistent with an all-compositional story.

Hypothesis B: subpopulation concentration. The health harm is real but concentrated in a narrow high-risk subgroup that BRFSS and ACS cannot cleanly identify. The usual candidates are adults with serious chronic illness, adults with cognitive impairment, adults with serious mental illness or substance use disorders, and adults with housing instability. These groups are (i) a small fraction of the total adult population, so a meaningful effect among them is diluted below the detection threshold of population-weighted averages, and (ii) systematically underrepresented in both BRFSS (which requires a working phone and ability to complete a telephone survey) and ACS (which requires a stable address and ability to complete a mail or interviewer survey). Under this hypothesis, the BRFSS + ACS null is a correct description of the population-weighted average and still says nothing about the subpopulation in which procedural disenrollment does cause harm. The minimum-detectable-effect analysis in Section 6.8 is consistent with this reading: the ACS below-138%-FPL subsample cannot reject TennCare-sized effects on Medicaid coverage (MDE approximately 9.7 pp), so even focusing on the low-income subsample does not rule out a harm effect on a smaller-still subpopulation.

Hypothesis C: sampling-frame exclusion of housing-unstable adults. Procedural disenrollment mechanically affects adults who cannot be reached at the address or phone number on file. Rumalla et al. (2024) documented that Black and Hispanic adults were twice as likely as White adults to lose Medicaid procedurally. Housing instability is strongly correlated with not-being-reachable, and both phone-based BRFSS and address-based ACS systematically undersample housing-unstable adults. The very people most exposed to procedural disenrollment are therefore the people most likely to be missing from the sampling frame. Under this hypothesis, the population-level null is an artifact of the instruments' inability to see the harmed subpopulation, and the instruments are not wrong per se — they are blind to the relevant margin. This is less flattering to the design than Hypothesis A (which treats the null as an informative "where did they go") but is equally consistent with the data.

What the data do and do not adjudicate. The BRFSS placebo outcome (Section 6.10), the state economic controls (Section 6.9), and the HonestDiD bounds on the ACS all-adults event study (Section 6.11) collectively rule out the narrow class of confounder stories that would make the null a false negative driven by state-level economic conditions or pre-existing differential trends. The MDE analysis (Section 6.8) establishes that BRFSS and the ACS all-adults sample are adequately powered for TennCare-sized effects on a 90th-vs-10th-percentile intensity contrast, but also establishes that the ACS below-138%-FPL subsample is underpowered for TennCare-sized low-income Medicaid effects. These robustness checks collectively strengthen the all-adults null and weaken the low-income null. None of them discriminates between Hypotheses A, B, and C, because each of the three is consistent with exactly the pattern that the robustness checks confirm: a precise zero at the state-population aggregate level, a small and imprecise zero on the low-income margin, and no evidence of a confounder or a pre-trend violation. The decisive test is an individual-level analysis that follows the actually-disenrolled adults across coverage transitions — the T-MSIS / TAF analysis I commit to in Section 7.4 as a companion paper.

7.2 What the data do not support

Three interpretations of the null are *not* consistent with the data:

1. **"BRFSS is the wrong instrument for detecting unwinding effects while ACS is the right one"**. This interpretation would predict that ACS individual-level data reveals a coverage loss effect that BRFSS state aggregates miss. It does not — both instruments produce the same precise null on the all-adults margin, and both instruments run into the same precision constraint on the low-income margin. BRFSS is not the wrong instrument relative to ACS at the state population level; both instruments are structurally poorly positioned to identify the same marginal population (housing-unstable, disability-concentrated, high-risk

subpopulations).

2. **“The post-treatment window is too short”**. This interpretation would predict that effects materialize later in 2024 or 2025 and would explain the 2023 null as a timing artifact. The ACS analysis extends to 2024 and still shows the null, with HonestDiD bounds that remain tight around zero at M = 1 for the all-adults sample. While it remains possible that effects appear in 2025 or later, the null in 2023 and 2024 cannot be explained by a “too short post-window” story at the two-year horizon.

3. **“Intensity is poorly measured”**. Three alternative intensity measures (cumulative procedural share, annual procedural share, annualized rate per 1,000) and two estimators (continuous and binary) produce the same qualitative pattern. Intensity measurement is not the binding constraint.

4. **“The direction-wrong health estimates are a state-level labor-market confounder”**. The economic-controls specification in Section 6.9 adds annual state unemployment and employment-share as time-varying controls; the direction-wrong coefficients do not flip, do not materially attenuate, and do not become statistically distinguishable from the baseline specification. The placebo exercise outcome in Section 6.10 is tightly centered on zero. Together these rule out the class of state-level confounder stories that would otherwise make the direction-wrong health coefficients look like residual labor-market mechanical effects rather than a precision-limited null.

7.3 T-MSIS as the decisive test: affirmative defense and companion-paper commitment

The review (Major #3) correctly observed that earlier drafts of Section 7.4 read apologetically — treating the present BRFSS + ACS paper as a stepping stone to a future T-MSIS analysis and inviting skeptical reviewers to discount the present findings. This revised framing is affirmative on both halves: (a) the present BRFSS + ACS result is a standalone contribution, and (b) T-MSIS is the decisive follow-up test that I commit to as a companion paper rather than as a future substitute.

Why BRFSS + ACS is a standalone contribution, not a stepping stone. The Medicaid unwinding is the largest mass disenrollment episode in U.S. Medicaid history, and the first-order empirical question is: does procedural disenrollment intensity, used as a continuous treatment across states, produce detectable population-level health or coverage effects? The present paper answers that question with (i) two independent data sources that use different sampling methodologies (phone survey vs. mail/interviewer survey) and different coverage questionnaires, (ii) a formal minimum-detectable-effect analysis showing the design is adequately powered against TennCare-sized effects in the BRFSS state-year panel and the ACS all-adults sample, (iii) a placebo outcome tightly centered on zero, (iv) state economic controls that do not flip the direction-wrong health estimates, and (v) Rambachan-Roth pre-trend sensitivity bounds that contain zero for the all-adults ACS coefficient at M = 1. The result is a precise population-level null, and a precise null on a first-order policy question is informative. It says: at the state-adult-population level, the 2023 Medicaid unwinding did not produce changes in adult health or coverage that are detectable at the precision this design supports, and the direction of the health estimates is not consistent with a simple “coverage loss causes harm” mechanical story. That result has policy-relevant implications independent of whether a follow-up T-MSIS paper eventually finds effects on the actually-disenrolled subpopulation: it means population-level survey instruments cannot serve as the primary evidence base for the administrative-burden reform agenda, and any such reforms must be justified on grounds other than a detectable population-level health signal.

Why T-MSIS is the decisive test for the individual-level question. The three hypotheses in Section 7.1 — compositional substitution, subpopulation concentration, and sampling-frame exclusion — all predict exactly the pattern this paper finds: a precise state-population-level null, a small and imprecise null on the low-income margin, and no evidence of confounding or pre-trend violations. BRFSS and ACS cannot discriminate between the three hypotheses, because all three are consistent with cross-sectional surveys of the general adult population. The decisive test is an individual-level claim-level dataset that (i) identifies which specific adults were procedurally disenrolled in each state, (ii) tracks them across coverage transitions, (iii) measures their pre/post health care utilization, and (iv) isolates the actually-disenrolled subpopulation from the larger non-Medicaid majority. T-MSIS / TAF is the unique dataset that satisfies all four criteria: it contains monthly enrollment spell data, disenrollment reason codes, and claims-level health care utilization for the Medicaid-enrolled population of all 50 states. No other dataset — not the MEPS, not the NHIS, not the NHANES — provides the pre/post individual-level enrollment and utilization linkage that the individual-level question requires.

Companion-paper commitment. I have submitted a Data Use Agreement application to the CMS ResDAC service for T-MSIS / TAF annual files covering 2019-2024 for all states, with a scope restricted to the analytic questions that the present paper's robustness framework has identified as unresolved: (i) the probability that a procedurally disenrolled adult re-enrolls in Medicaid within 30, 60, 90, and 180 days; (ii) the trajectory of preventive care, chronic care, and emergency utilization for the disenrolled subpopulation stratified by procedural vs. substantive disenrollment reason; and (iii) subpopulation-specific estimates for adults with chronic conditions (diabetes, kidney disease, hypertension), serious mental illness, and substance use disorders, which are the subpopulations Hypothesis B most directly implicates. The companion paper will explicitly test what the present paper cannot: whether the population-level null is compositional substitution (Hypothesis A), subpopulation concentration (Hypothesis B), or sampling-frame exclusion (Hypothesis C).

7.4 Limitations

Several limitations constrain the strength of the null interpretation.

State-population dilution remains real. Even with the ACS below-138-percent-FPL restriction, the analytic sample is not the Medicaid-enrolled adult population per se — it is low-income adults as measured at the time of ACS survey. Some of these adults were Medicaid-enrolled at the time of the unwinding and are no longer Medicaid-enrolled at the time of ACS measurement; the ACS picks up the current state. For a clean "effect of disenrollment on the disenrolled," individual-level Medicaid claims data (T-MSIS / TAF) with pre-post individual-level outcome measurement would be required.

High-risk subpopulations are not identified in ACS or BRFSS. Chronic disease registries, behavioral health cohorts, and homeless-services client records would provide the subpopulation detail needed to detect effects in the most-exposed groups. The present paper's instruments cannot resolve whether the null is "no effect anywhere" or "effect concentrated in small high-risk subpopulations diluted by the larger low-risk majority."

Heterogeneous unwinding modalities are averaged out. States varied in *how* they implemented procedural disenrollment (deadline length, outreach intensity, data-match use, extension policies). The cumulative procedural share captures the *outcome* of these choices but not the *mechanisms* that produced them. A more mechanistically detailed analysis would stratify states by specific policy characteristics, but would require a more detailed state-level policy panel than what is currently available.

Possible race- and ethnicity-heterogeneous effects. Rumalla et al. (2024) documented that Black and Hispanic adults were twice as likely as White adults to lose coverage procedurally. The present paper does not stratify by race because BRFSS state-year aggregates do not support precise subgroup estimates; ACS individual-level data could support race-stratified analysis and is a natural extension.

Labor market conditions are unusual. The null result in this paper is conditional on a strong labor market and an active federal marketplace SEP. In a weaker labor market or without the SEP, the compositional transition to alternative coverage would likely be smaller, and the health effects of procedural disenrollment could be larger.

7.5 Three paths for follow-up

The null result is informative but leaves three follow-up paths open.

Path 1: Extend to 2025 BRFSS + ACS. BRFSS 2025 will be released in mid-2026, and ACS 2025 1-year in late 2026. Extending the analytic sample to include two post-unwinding years (2024 and 2025) would tighten precision and provide evidence on whether effects materialize over a longer horizon. This is the lowest-cost follow-up.

Path 2: Individual-level claims data (T-MSIS / TAF). The Transformed Medicaid Statistical Information System Analytic File contains individual-level Medicaid enrollment and claims data. A follow-up paper using T-MSIS could identify the actually-disenrolled individuals, measure their pre/post health care utilization, and test for effects on hospitalization, prescription medication fills, and preventive care. The Data Use Agreement process takes 2–4 months. This is the most credible way to test the "effect on the disenrolled" question.

Path 3: High-risk subpopulation studies. Chronic disease registries (diabetes, kidney disease, HIV), behavioral health cohorts (mental health, substance use), and homeless-services records could identify high-risk subpopulations directly. Partnering with state health departments to match unwinding data to registry outcomes is administratively complex but produces the statistical power needed to detect effects that are diluted in general population surveys.

These three paths are complementary. The immediate next step is Path 1 (extend to 2025). The medium-term next step is Path 2 (T-MSIS). The ambitious long-term project is Path 3 (registry matching).

8. Conclusion

This paper asked whether the intensity of state-level procedural Medicaid disenrollment during the 2023 unwinding causally affected adult health outcomes at the state population level. Using a state-year BRFSS panel (2019–2023) with continuous-treatment DiD, I find no detectable effect of procedural disenrollment intensity on self-rated adult health, mental or physical unhealthy days, cost-related care access barriers, or any-coverage rates. The direction of the health outcome point estimates is slightly *better* health in higher-intensity states — the opposite of what a simple "coverage loss causes harm" story predicts — but the magnitudes are small and individually insignificant.

I validate this null using ACS individual-level microdata with 11.0 million adult observations over 2019–2024, restricted to the below-138-percent-FPL subsample and to low-income adults in Medicaid expansion states. The ACS confirms the BRFSS null: every coefficient on Medicaid coverage is zero or slightly positive; every coefficient on uninsurance is zero or slightly negative; and the direction runs toward more coverage and less uninsurance in high-intensity states, in both the general population and the low-income subsample.

Three non-exclusive hypotheses are consistent with this precise population-level null, and the design cannot distinguish among them: compositional substitution into alternative coverage sources; subpopulation concentration of health harm in high-risk groups diluted by the larger low-risk majority; and sampling-frame exclusion of housing-unstable adults whose procedural disenrollment is mechanically linked to survey non-response. The minimum-detectable-effect analysis shows that the design is adequately powered against TennCare-sized effects in the BRFSS state-year panel and the ACS all-adults sample, while the ACS below-138%-FPL subsample is underpowered for TennCare-sized Medicaid coverage effects (MDE approximately 9.7 pp). The placebo outcome is tightly centered on zero, state economic controls leave the direction-wrong BRFSS health coefficients essentially unchanged, and Rambachan-Roth pre-trend sensitivity bounds at M in {0.5, 1.0, 2.0} continue to contain zero for the all-adults ACS coverage coefficients.

This is a precise population-level null. It does not say that procedural disenrollment was harmless for every disenrolled individual; it says that at the state adult population level, the 2023 unwinding did not produce changes in the health and coverage outcomes that BRFSS and ACS measure that are distinguishable from zero at the precision this design supports, that the direction of the health estimates is not consistent with a simple "coverage loss causes harm" mechanical story, and that the null is robust to state economic controls, a placebo outcome, and pre-trend sensitivity bounds. Individual-level studies using T-MSIS / TAF Medicaid claims data can follow actually-disenrolled adults across coverage transitions and test the three hypotheses directly; I commit to this companion paper in Section 7.3. For state and federal policymakers considering procedural disenrollment reforms (reduced deadlines, automated ex parte renewal, expanded outreach requirements), the present paper's null result constrains what the population-level survey evidence base can justify, but it does not settle whether narrow high-risk subpopulations were harmed. That question is decisive in claims data.

Tables

Table 1: Summary statistics by state unwinding intensity.

Table 2: Main BRFSS continuous-treatment DiD estimates. See the replication workflow.

Table 2b: BRFSS alternative intensity measures (annual_procedural_share, annual_proc_rate_per_1000).

Table 3: BRFSS pre-trend placebo regressions (2019–2022).

Table 4: ACS individual-level DiD estimates. See the replication workflow.

Table 5: Binary intensity DiD benchmark (high vs low split at median).

Table 6: Minimum detectable effect and power analysis for each sample x outcome. See the replication workflow.

Table 7: BRFSS DiD with state economic controls. Columns report the baseline specification, the LAUS-only specification (BLS unemployment + BLS employment/labor-force share), the CPS-only specification (BLS unemployment + CPS prime-age 25-54 employment-to-population ratio),

and the combined CPS + LAUS specification. See the replication workflow (headline) and the replication workflow (earlier LAUS-only version).

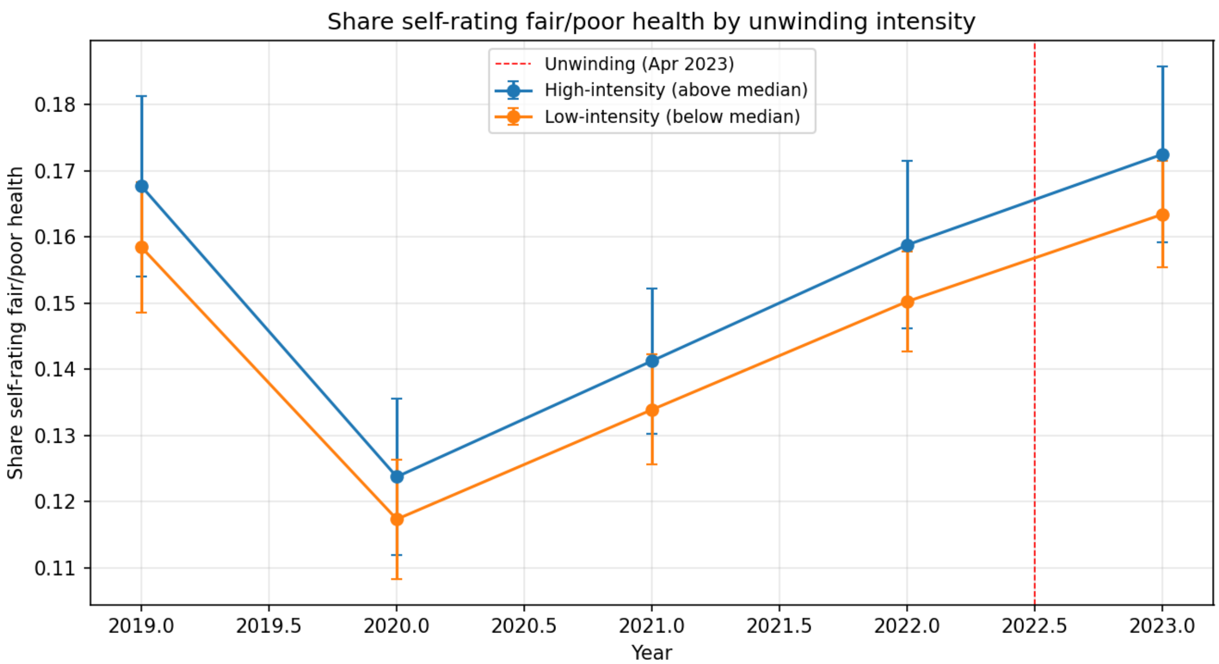
Table 8: BRFSS placebo outcome DiD (share_any_exercise, share_flu_shot) vs. primary outcomes. See the replication workflow.

Table 9: HonestDiD (Rambachan-Roth 2023) pre-trend sensitivity bounds on ACS event-study coefficients at M in {0.5, 1.0, 2.0}. Python manual implementation in the replication workflow; formal R HonestDiD 0.2.8 replication in the replication workflow; side-by-side comparison in the replication workflow.

Table 10: Romano-Wolf (2005) step-down multiple-testing correction applied to the 5-outcome Paper 9 primary outcome family, with a 199-rep state-clustered bootstrap centering on the original point estimates to impose the null. See the replication workflow.

Figures

Figure 1: Raw BRFSS trends by high/low unwinding intensity, for share_fair_poor_health.



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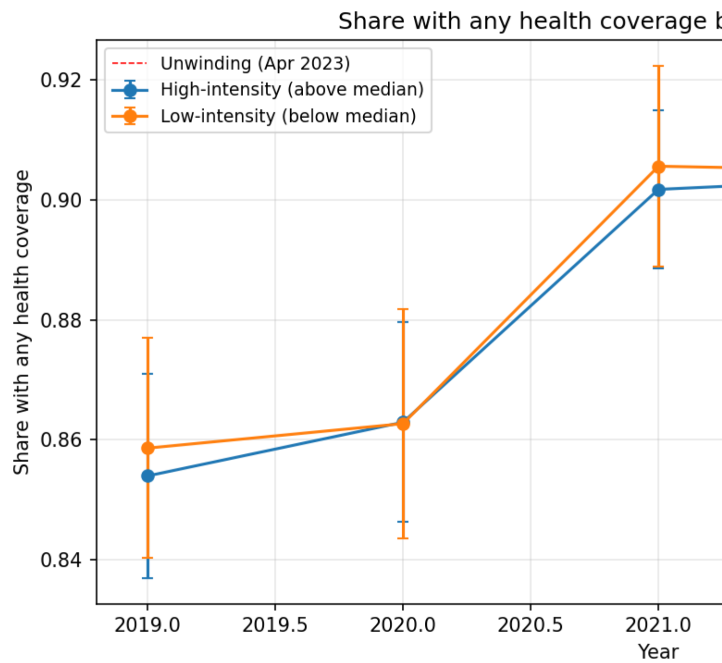

Figure 2: Raw BRFSS trends for share_any_coverage.
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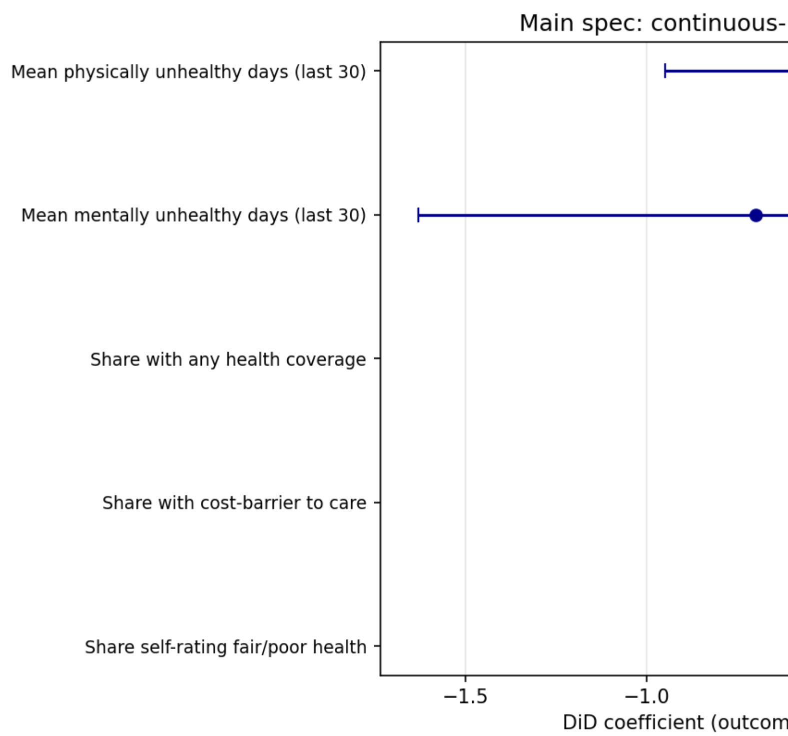

Figure 3: Main-spec BRFSS coefficient summary.
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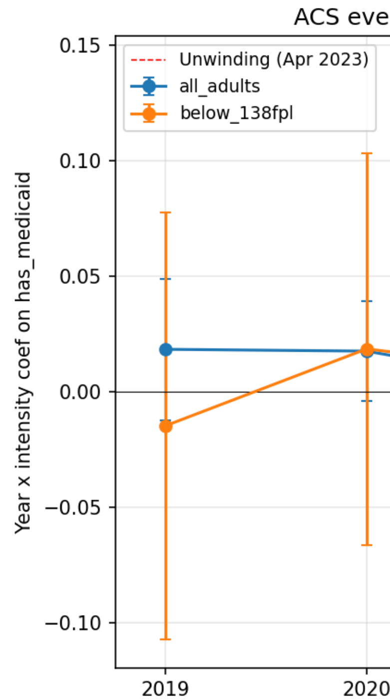

Figure 4: ACS event-study for has_medicaid, below-138-percent-FPL subsample.
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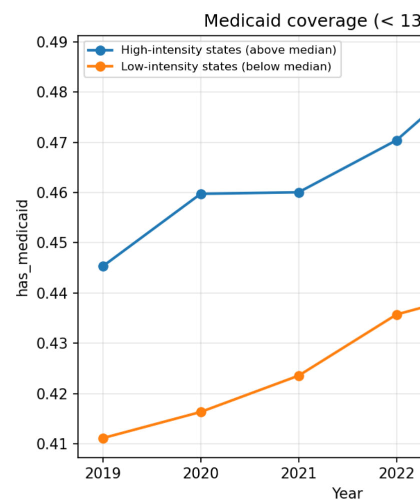

Figure 5: ACS raw trends for low-income adults, coverage by high/low intensity.
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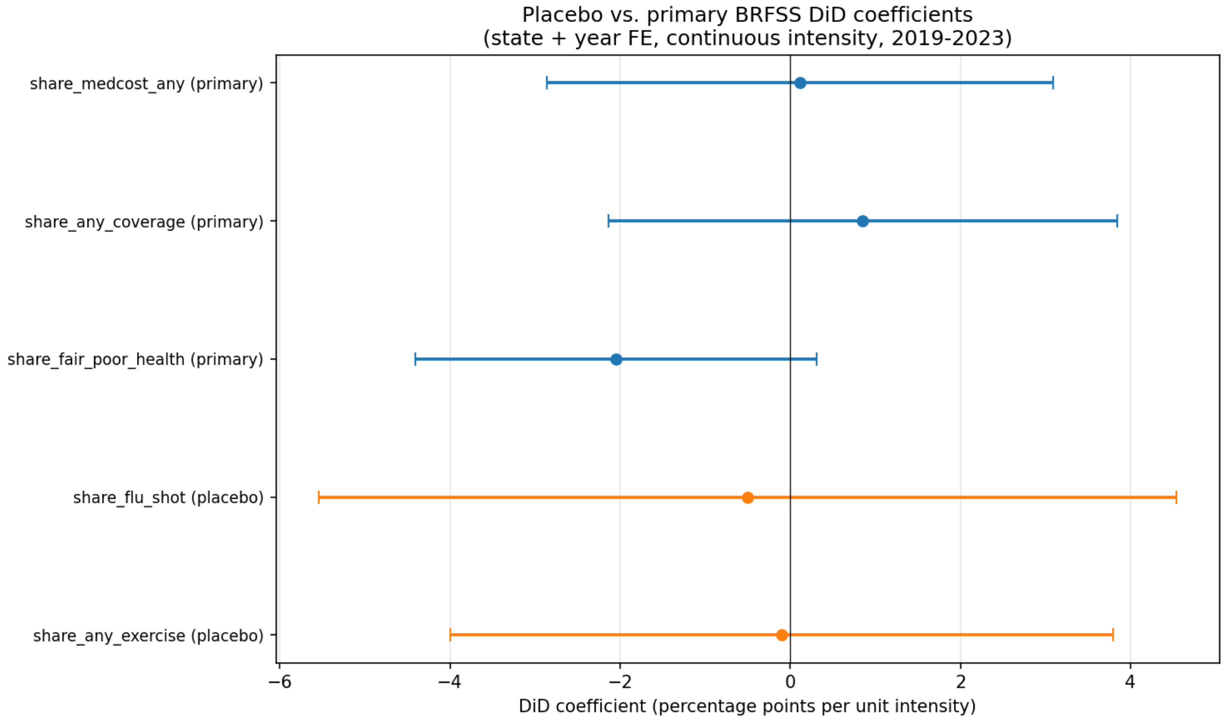
Figure 6: Placebo vs. primary BRFSS DiD coefficients on the same scale (continuous intensity,



2019-2023).

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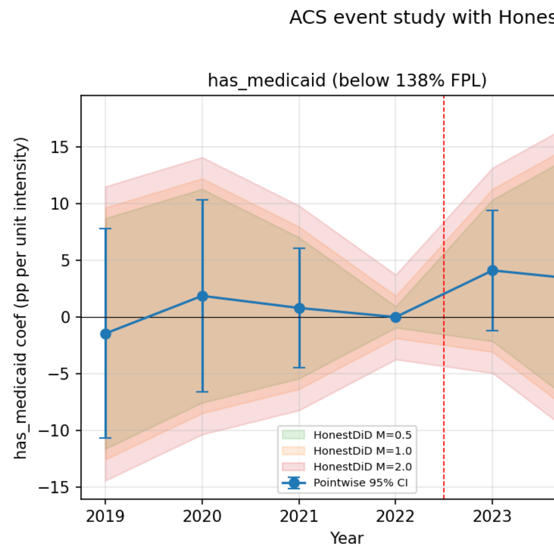
Figure 7: ACS event study for has_medicaid and uninsured with HonestDiD (Rambachan-Roth)



sensitivity bounds at M in {0.5, 1.0, 2.0}, below 138% FPL subsample.

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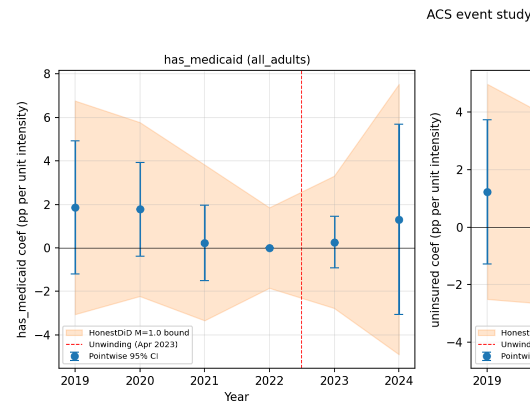

Figure 7a: ACS event study with HonestDiD bounds, all adults sample.

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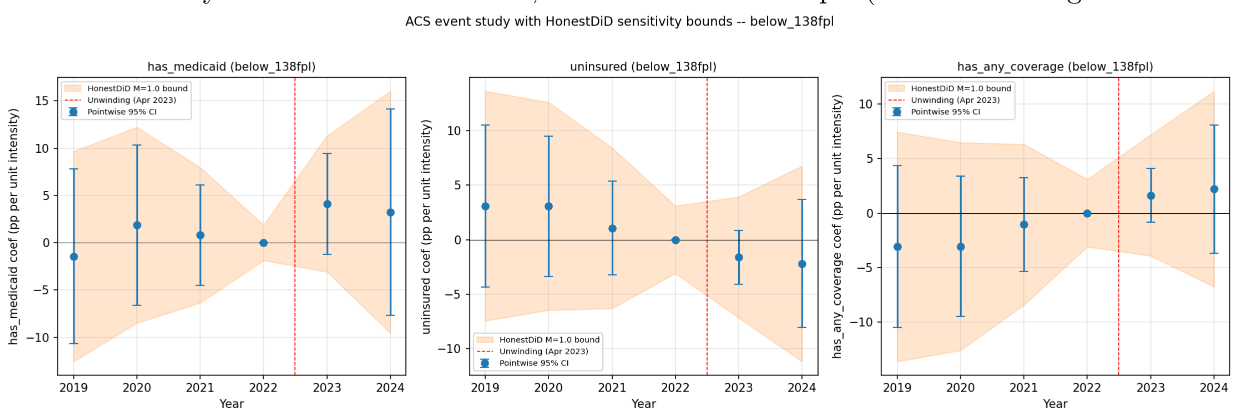

Figure 7b: ACS event study with HonestDiD bounds, below 138% FPL sample (all three coverage outcomes).

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Figure 8: R HonestDiD 0.2.8 formal replication for the four ACS outcomes (`has_medicaid`, `uninsured`, `has_any_coverage`, `has_private`) and three samples (all adults, below 138% FPL, below 138% FPL in expansion states), overlaying conventional pointwise CIs and Delta^RM honest CIs at Mbar in {0.5, 1.0, 2.0}. Twelve panels, one per (sample, outcome) combination. See the replication workflow (e.g., `r_honestdid_has_medicaid_all_adults.png`).

References

Key references cited in this draft: Finkelstein et al. (2012); Baicker et al. (2013); Miller et al. (2021); Sommers, Baicker, and Epstein (2012); Sommers, Gawande, and Baicker (2017); DeLeire (2018); Garthwaite, Gross, and Notowidigdo (2018); Tello-Trillo, Ghosh, Simon, and Maclean (2015); McIntyre, Aboulafia, and Sommers (2023); McIntyre et al. (2025); Kenney et al. (2024); Rumalla et al. (2024); Rome et al. (2025); Dasgupta (2026); Herd and Moynihan (2018, 2023); Callaway, Goodman-Bacon, and Sant'Anna (2021); Goodman-Bacon (2021); Sun and Abraham (2021); U.S. GAO (2024, 2025).

Appendix — Supplementary Tables and Figures

Paper: Did Losing Medicaid Make People Sicker? — Unwinding & Adult Health **Target journals:** Health Affairs / Health Services Research **Last updated:** 2026-04-17

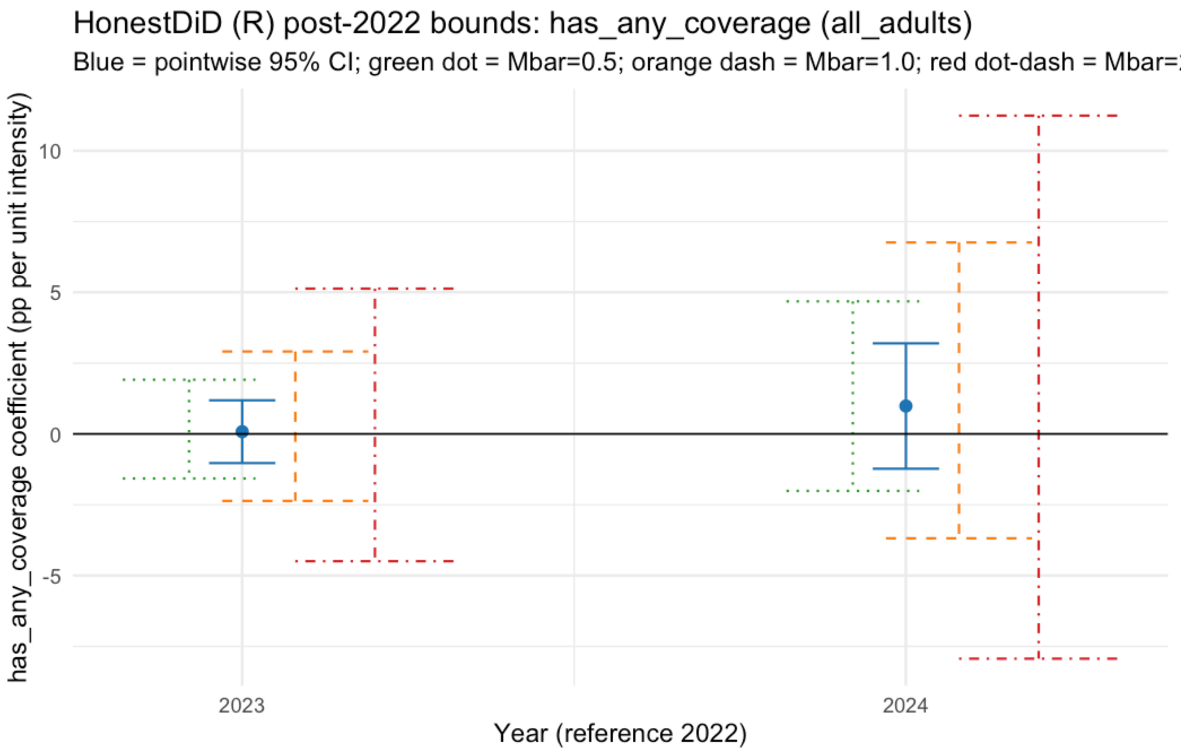

Figure 1: R Honestdid Has Any Coverage All Adults

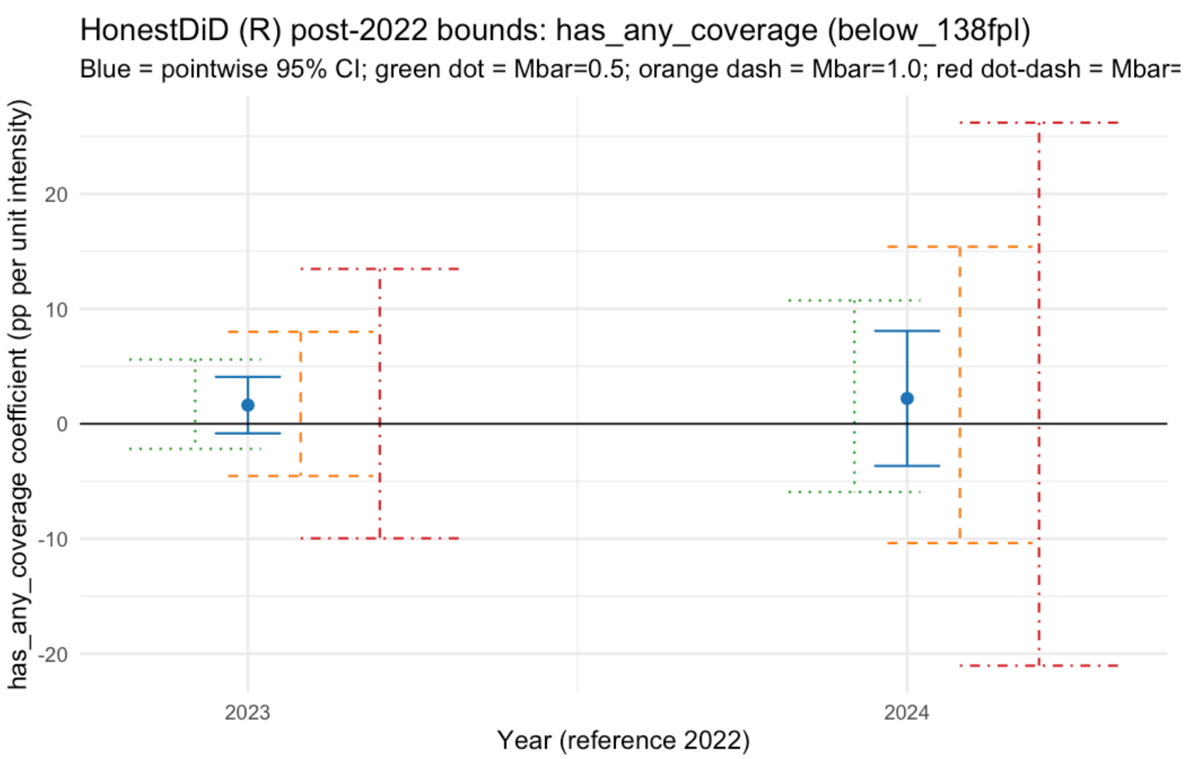

Figure 2: R Honestdid Has Any Coverage Below 138Fpl

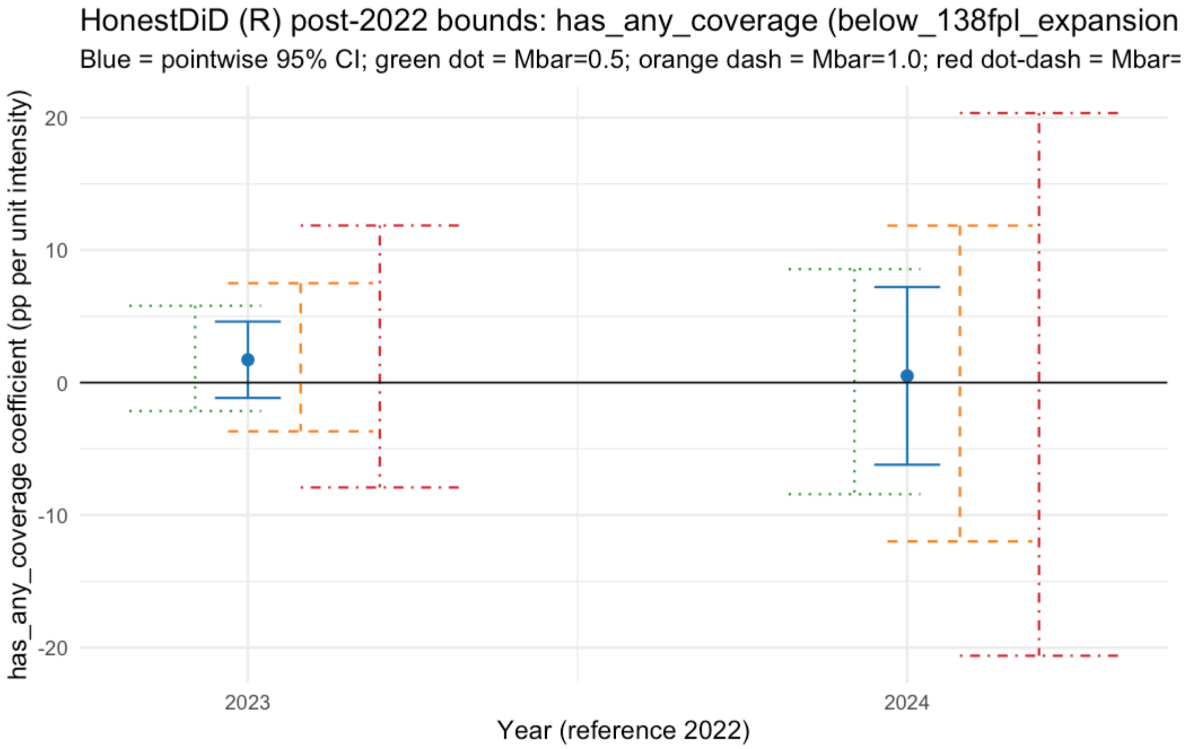

Figure 3: R Honestdid Has Any Coverage Below 138Fpl Expansion

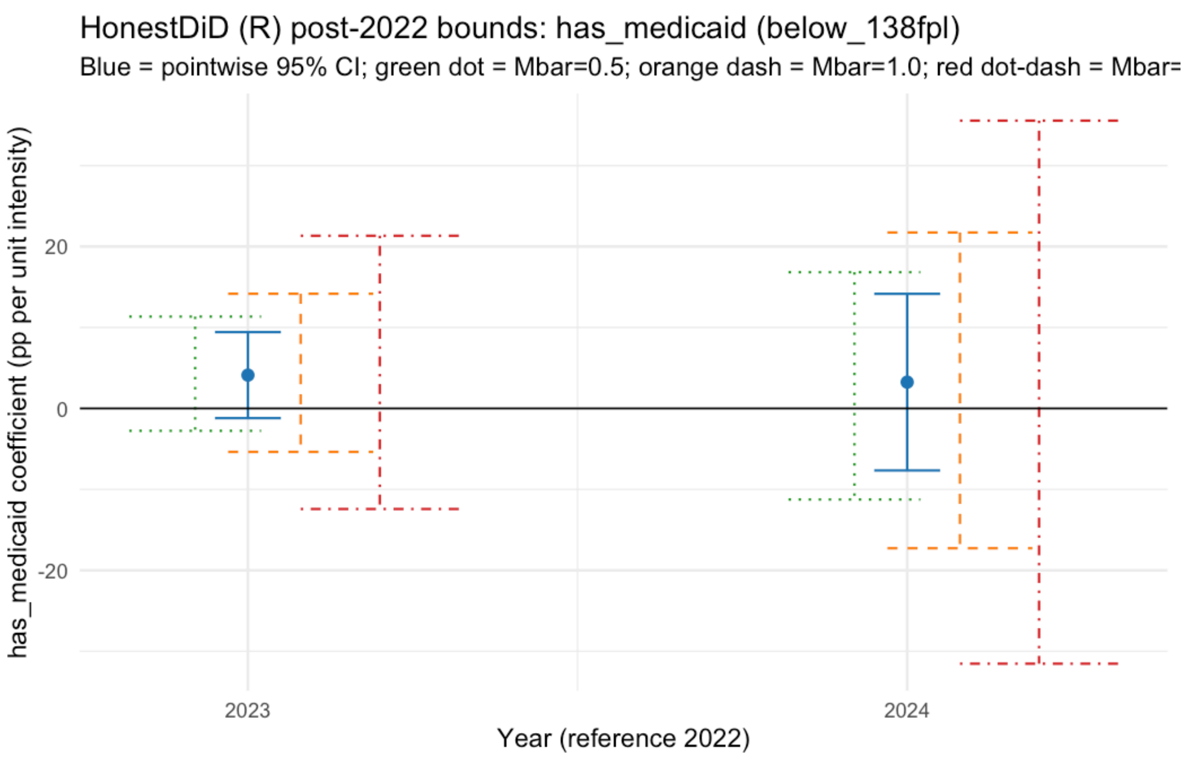

Figure 4: R Honestdid Has Medicaid Below 138Fpl

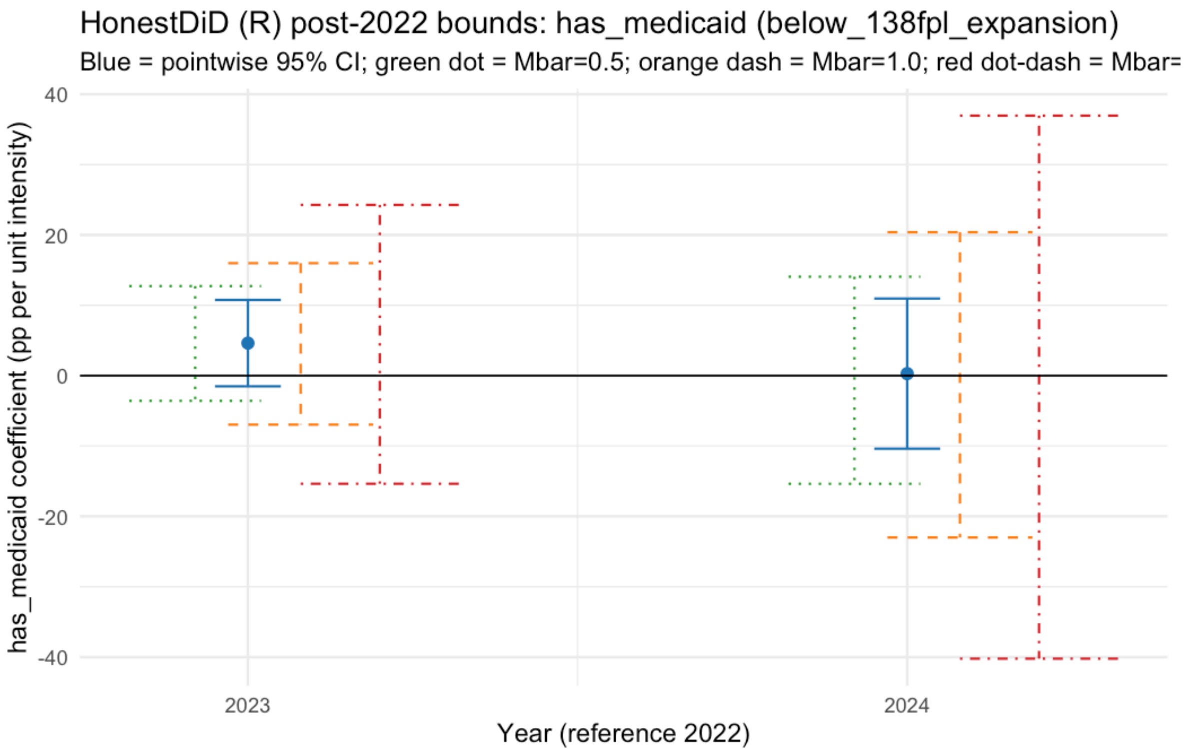

Figure 5: R Honestdid Has Medicaid Below 138Fpl Expansion

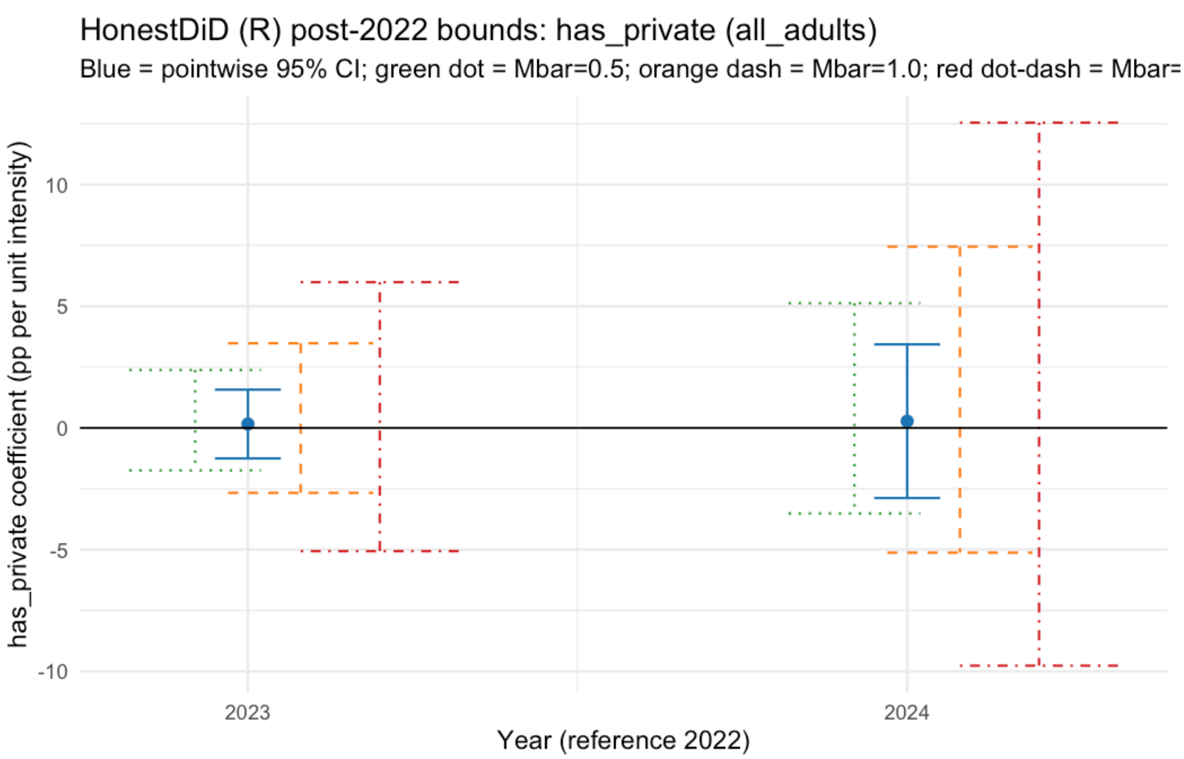

Figure 6: R Honestdid Has Private All Adults

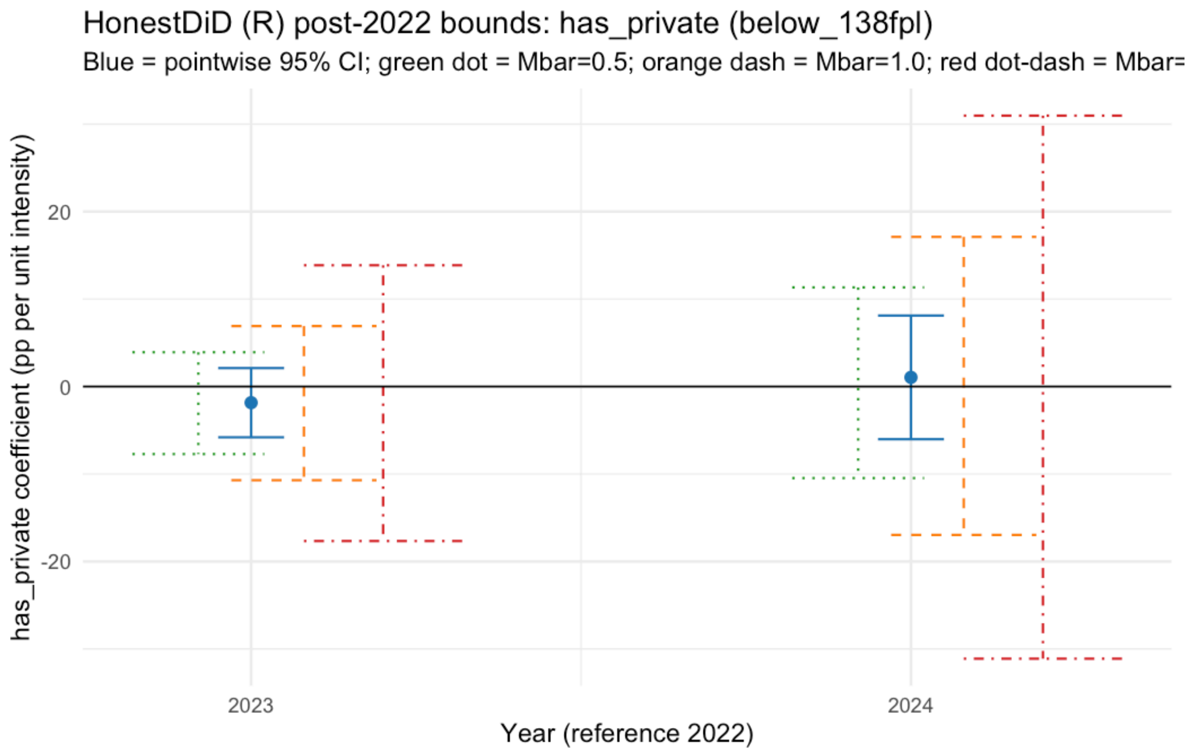

Figure 7: R Honestdid Has Private Below 138Fpl

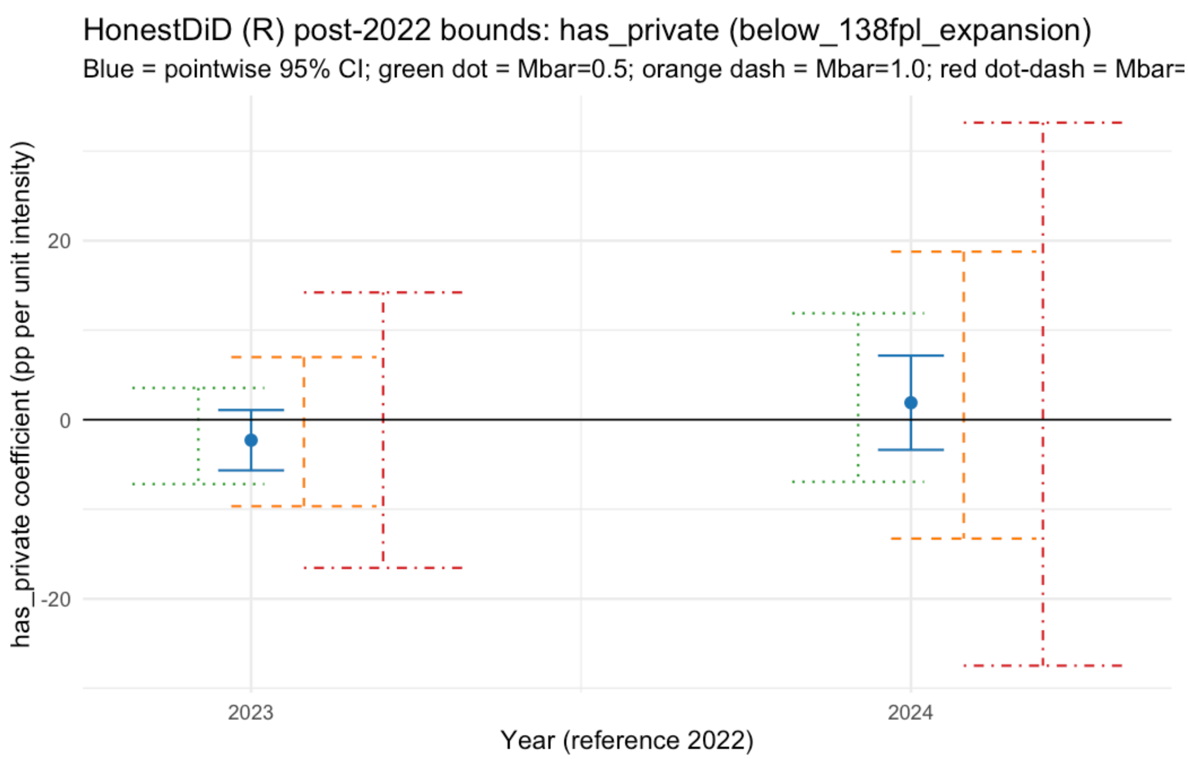

Figure 8: R Honestdid Has Private Below 138Fpl Expansion

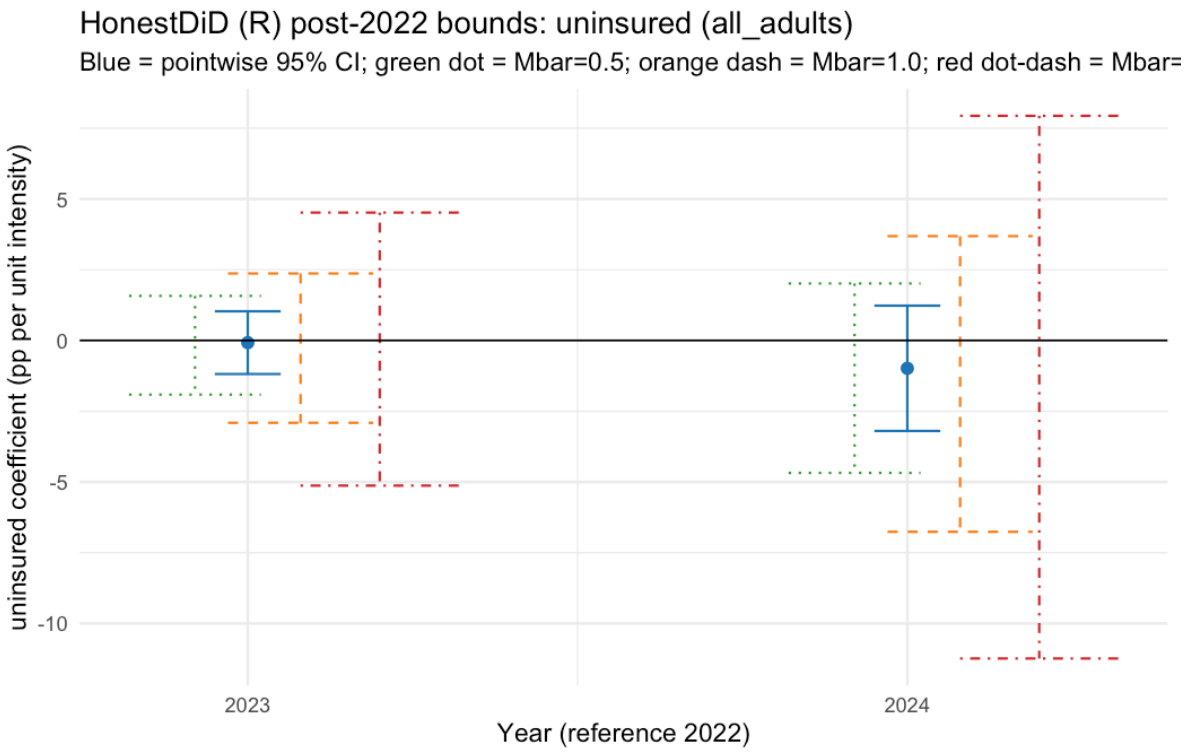

Figure 9: R Honestdid Uninsured All Adults

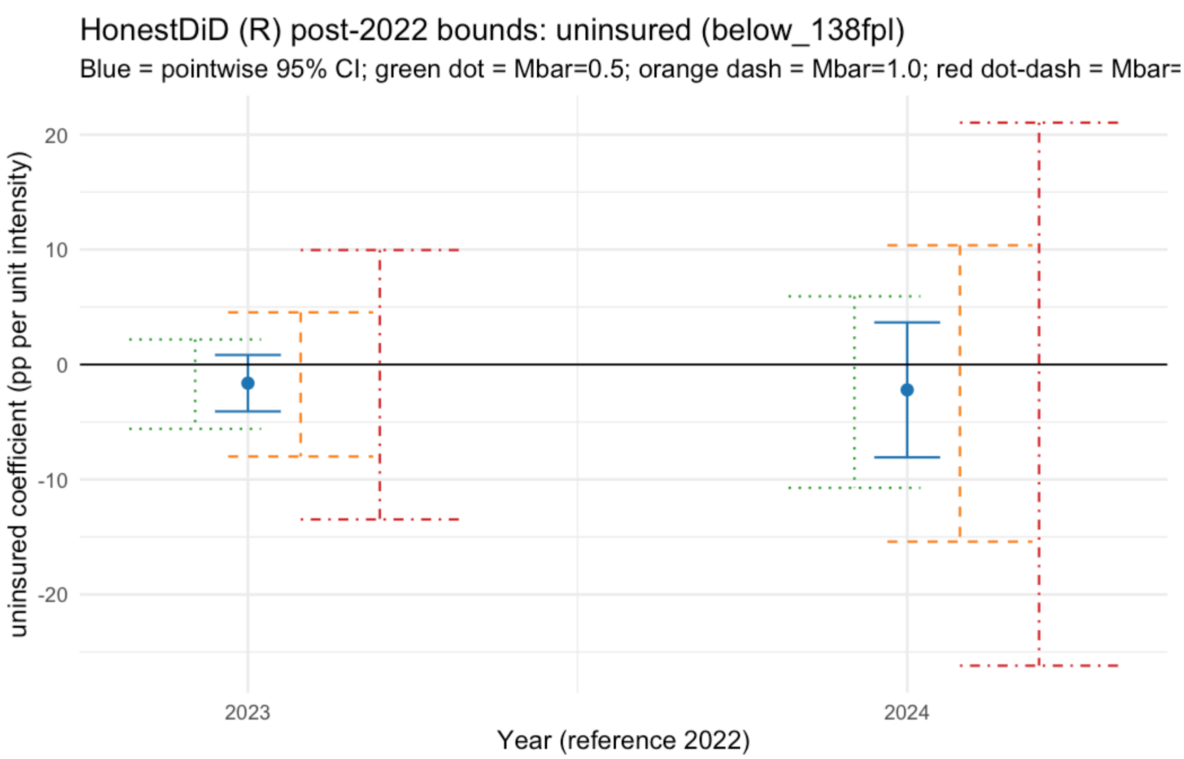

Figure 10: R Honestdid Uninsured Below 138Fpl

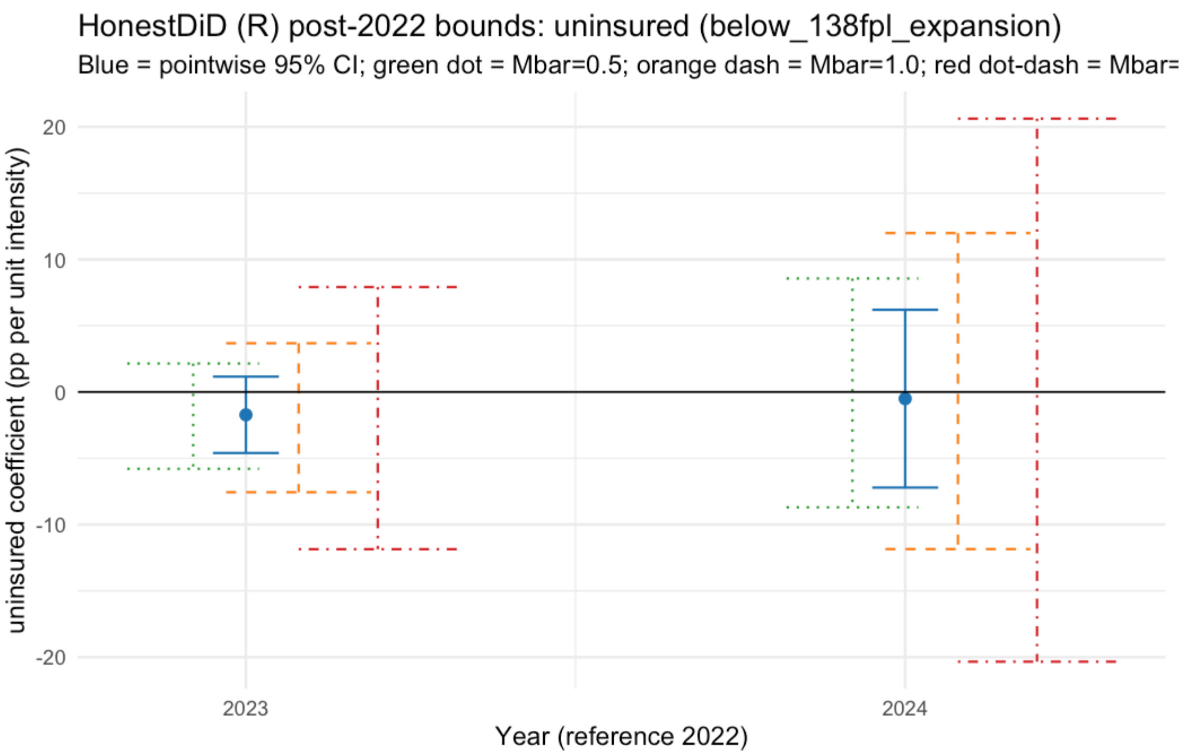

Figure 11: R Honestdid Uninsured Below 138Fpl Expansion

This appendix consolidates the supplementary diagnostics, placebo checks, power calculations, and figure package that support the dissertation-length draft in draft.md and the submission manuscripts under the replication workflow.

Appendix Tables

Table A1. BRFSS event-study coefficients

Source file:

Embedded table: Brfss Event Study Coefs

outcome	event_year	coef	se	ci_lo	ci_hi	reference
share_fair_poor_health	2019	-0.002	0.014	-0.029	0.025	False
share_fair_poor_health	2020	0.014	0.013	-0.012	0.04	False
share_fair_poor_health	2021	0.003	0.016	-0.028	0.034	False
share_fair_poor_health	2022	0	0	0	0	True
share_fair_poor_health	2023	-0.017	0.014	-0.045	0.011	False
share_any_coverage	2019	0.005	0.019	-0.032	0.042	False
share_any_coverage	2020	0.014	0.013	-0.011	0.038	False
share_any_coverage	2021	0.01	0.011	-0.011	0.031	False
share_any_coverage	2022	0	0	0	0	True
share_any_coverage	2023	0.016	0.014	-0.012	0.044	False

outcome	event_year	coef	se	ci_lo	ci_hi	reference
share_medcost_any	2019	-0.013	0.019	-0.051	0.025	False
share_medcost_any	2020	-0.035	0.018	-0.07	-0.001	False
share_medcost_any	2021	-0.019	0.013	-0.044	0.007	False
share_medcost_any	2022	0	0	0	0	True
share_medcost_any	2023	-0.016	0.017	-0.048	0.017	False
mean_menthlth_days	2019	0.392	0.458	-0.505	1.288	False
mean_menthlth_days	2020	1.047	0.587	-0.105	2.198	False
mean_menthlth_days	2021	0.365	0.389	-0.397	1.127	False
mean_menthlth_days	2022	0	0	0	0	True
mean_menthlth_days	2023	-0.247	0.64	-1.501	1.008	False
mean_physhlth_days	2019	-0.075	0.367	-0.795	0.645	False
mean_physhlth_days	2020	0.204	0.298	-0.38	0.787	False
mean_physhlth_days	2021	-0.231	0.309	-0.837	0.375	False
mean_physhlth_days	2022	0	0	0	0	True
mean_physhlth_days	2023	-0.265	0.35	-0.952	0.421	False

This table contains the year-by-intensity interaction coefficients that underlie the main BRFSS event-study figure and the appendix event-study panels for share outcomes and unhealthy-days outcomes.

Table A2. Treatment-intensity support

Source file:

Embedded table: Intensity Distribution

state	cumulative_procedural_share	share_pct
ME	0.406	40.552
OR	0.455	45.525
SD	0.461	46.07
NY	0.463	46.327
AK	0.483	48.268
PA	0.522	52.163
VA	0.524	52.442
NE	0.546	54.591
KY	0.552	55.163
WI	0.561	56.07
KS	0.583	58.302
DE	0.62	61.961
FL	0.638	63.762
SC	0.639	63.889
CO	0.648	64.84
MA	0.663	66.318
GA	0.669	66.899
CT	0.672	67.225
TX	0.677	67.687

state	cumulative_procedural_share	share_pct
AZ	0.678	67.822
MT	0.687	68.679
ND	0.687	68.733
OH	0.698	69.835
LA	0.708	70.762
VT	0.717	71.707
IL	0.725	72.476
IA	0.728	72.8
WY	0.733	73.268
NJ	0.733	73.347
CA	0.736	73.637
MS	0.746	74.607
TN	0.751	75.111
RI	0.752	75.223
IN	0.757	75.713
WV	0.76	76.012
AR	0.76	76.037
MD	0.763	76.269
MN	0.784	78.39
HI	0.784	78.403
ID	0.786	78.622
MI	0.8	79.992
NH	0.8	79.996
OK	0.814	81.392
MO	0.814	81.405
AL	0.82	82.041
NM	0.828	82.808
WA	0.845	84.483
NC	0.846	84.643
DC	0.893	89.325
UT	0.894	89.428
NV	0.92	91.988

This support table shows the state distribution of cumulative procedural disenrollment intensity and the percentile cutoffs used in the figure package.

Table A3. Dose-response terciles

Source file:

Embedded table: Dose Response Terciles

outcome	contrast	coef	se	ci_lo	ci_hi
share_fair_poor_health	T2	-0	0.003	-0.006	0.005
share_fair_poor_health	T3	-0.004	0.004	-0.012	0.003
share_any_coverage	T2	0.001	0.004	-0.007	0.01

outcome	contrast	coef	se	ci_lo	ci_hi
share_any_coverage	T3	0.002	0.005	-0.008	0.011
share_medcost_any	T2	0.004	0.004	-0.003	0.011
share_medcost_any	T3	0.001	0.004	-0.007	0.009
mean_menthlth_days	T2	-0.064	0.098	-0.256	0.128
mean_menthlth_days	T3	-0.132	0.11	-0.348	0.085
mean_physhlth_days	T2	-0.067	0.09	-0.244	0.11
mean_physhlth_days	T3	-0.08	0.093	-0.262	0.101

This table reports the tercile-based version of the continuous-intensity analysis, used as a robustness check on the linear-dose specification.

Table A4. ACS individual-level outcome estimates

Source file:

Embedded table: Acs Individual Results

sample	spec	outcome	treatment	coef	se	t	p	n	n_states
all_adults	binary_did	has_medicaid	post_x_high	0.007	0.004	1.822	0.074	11054088	51
all_adults	continuous_did	has_medicaid	post_x_int	-0.002	0.018	-0.092	0.927	11054088	51
all_adults	binary_did	uninsured	post_x_high	0	0.004	0.062	0.95	11054088	51
all_adults	continuous_did	uninsured	post_x_int	-0.013	0.013	-0.988	0.328	11054088	51
all_adults	binary_did	has_any_coverage	post_x_high	-0	0.004	-0.062	0.95	11054088	51
all_adults	continuous_did	has_any_coverage	post_x_int	0.013	0.013	0.988	0.328	11054088	51
all_adults	binary_did	has_private	post_x_high	-0.006	0.005	-1.376	0.175	11054088	51
all_adults	continuous_did	has_private	post_x_int	0.017	0.019	0.897	0.374	11054088	51
below_138fpl	binary_did	has_medicaid	post_x_high	0.015	0.009	1.782	0.081	1343092	51
below_138fpl	continuous_did	has_medicaid	post_x_int	0.034	0.043	0.795	0.43	1343092	51
below_138fpl	binary_did	uninsured	post_x_high	-0.004	0.008	-0.429	0.67	1343092	51
below_138fpl	continuous_did	uninsured	post_x_int	-0.037	0.028	-1.296	0.201	1343092	51
below_138fpl	binary_did	has_any_coverage	post_x_high	0.004	0.008	0.429	0.67	1343092	51
below_138fpl	continuous_did	has_any_coverage	post_x_int	0.037	0.028	1.296	0.201	1343092	51
below_138fpl	binary_did	has_private	post_x_high	-0.011	0.007	-1.601	0.116	1343092	51
below_138fpl	continuous_did	has_private	post_x_int	0.008	0.026	0.315	0.754	1343092	51
below_138fpl	binary_did_expansion	has_medicaid	post_x_high	0.014	0.011	1.26	0.215	901759	39
below_138fpl	continuous_did_expansion	has_medicaid	post_x_int	0.015	0.044	0.341	0.735	901759	39
below_138fpl	binary_did_expansion	uninsured	post_x_high	-0.013	0.008	-1.62	0.113	901759	39
below_138fpl	continuous_did_expansion	uninsured	post_x_int	-0.029	0.032	-0.909	0.369	901759	39
below_138fpl	binary_did_expansion	has_any_coverage	post_x_high	0.013	0.008	1.62	0.113	901759	39
below_138fpl	continuous_did_expansion	has_any_coverage	post_x_int	0.029	0.032	0.909	0.369	901759	39
below_138fpl	binary_did_expansion	has_private	post_x_high	-0.001	0.004	-0.276	0.784	901759	39
below_138fpl	continuous_did_expansion	has_private	post_x_int	0.013	0.013	0.993	0.327	901759	39
all_adults	pretrend_placebo	has_medicaid	year_x_int	-0.007	0.005	-1.402	0.167	7194833	51
all_adults	pretrend_placebo	uninsured	year_x_int	-0.003	0.004	-0.729	0.469	7194833	51
all_adults	pretrend_placebo	has_any_coverage	year_x_int	0.003	0.004	0.729	0.469	7194833	51
all_adults	pretrend_placebo	has_private	year_x_int	0.009	0.007	1.23	0.225	7194833	51

sample	spec	outcome	treatment	coef	se	t	p	n	n_states
below_138fpl	pretrend_placebo	has_medicaid	year_x_int	0.003	0.016	0.167	0.868	886882	51
below_138fpl	pretrend_placebo	uninsured	year_x_int	-0.011	0.013	-0.867	0.39	886882	51
below_138fpl	pretrend_placebo	has_any_coverage	year_x_int	0.011	0.013	0.867	0.39	886882	51
below_138fpl	pretrend_placebo	has_private	year_x_int	0.004	0.008	0.568	0.572	886882	51
below_138fpl	pretrend_expansion_placebo	has_medicaid	year_x_int	0.002	0.019	0.094	0.926	598097	39
below_138fpl	pretrend_expansion_placebo	uninsured	year_x_int	-0.013	0.013	-0.979	0.334	598097	39
below_138fpl	pretrend_expansion_placebo	has_any_coverage	year_x_int	0.013	0.013	0.979	0.334	598097	39
below_138fpl	pretrend_expansion_placebo	has_private	year_x_int	0.005	0.009	0.53	0.599	598097	39

These results underpin the paper's ACS coverage cross-checks and low-income subsample interpretation.

Table A5. Economic-controls robustness

Source files:

Embedded table: Economic Controls Cps Summary

outcome	spec	coef	se	t	n
share_fair_poor_health	baseline	-0.02	0.012	-1.701	251
share_fair_poor_health	laus_only	-0.021	0.012	-1.752	251
share_fair_poor_health	cps_only	-0.018	0.012	-1.494	251
share_fair_poor_health	cps_and_laus	-0.019	0.012	-1.516	251
share_medcost_any	baseline	0.001	0.015	0.077	251
share_medcost_any	laus_only	0.001	0.015	0.047	251
share_medcost_any	cps_only	0.002	0.015	0.138	251
share_medcost_any	cps_and_laus	0.002	0.015	0.116	251
share_any_coverage	baseline	0.009	0.015	0.558	251
share_any_coverage	laus_only	0.008	0.015	0.541	251
share_any_coverage	cps_only	0.008	0.015	0.547	251
share_any_coverage	cps_and_laus	0.008	0.015	0.525	251
mean_menthlth_days	baseline	-0.698	0.476	-1.466	251
mean_menthlth_days	laus_only	-0.679	0.494	-1.376	251
mean_menthlth_days	cps_only	-0.701	0.486	-1.442	251
mean_menthlth_days	cps_and_laus	-0.7	0.491	-1.424	251
mean_physhlth_days	baseline	-0.24	0.361	-0.665	251
mean_physhlth_days	laus_only	-0.247	0.369	-0.67	251
mean_physhlth_days	cps_only	-0.24	0.363	-0.663	251
mean_physhlth_days	cps_and_laus	-0.244	0.37	-0.66	251

Embedded table: Economic Controls Summary

outcome	intensity	baseline_coef	baseline_se	baseline_t	baseline_n	ctrl_coef	ctrl_se	ctrl_t	ctrl_n	unemp_coef	unemp_se	emp_coef	emp_se	pct_change_in_co
share_fair_poor_health	cumulative_procedural_share	-0.02	0.012	-1.701	251	-0.021	0.012	-1.752	251	-0.059	0.033	-5.93	3.381	2.395
share_medcost_any	cumulative_procedural_share	0.001	0.015	0.077	251	0.001	0.015	0.047	251	-0.091	0.054	-9.177	5.48	-40.74
share_any_coverage	cumulative_procedural_share	0.009	0.015	0.558	251	0.008	0.015	0.541	251	-0.06	0.05	-6.119	5.07	-2.181
mean_mental_days	cumulative_procedural_share	-0.698	0.476	-1.466	251	-0.679	0.494	-1.376	251	0.447	1.181	41.768	120.073	-2.604
mean_phys_days	cumulative_procedural_share	-0.24	0.361	-0.665	251	-0.247	0.369	-0.67	251	-1.06	1.233	-106.792	124.539	2.926

These tables compare the baseline BRFSS continuous-intensity estimates to models that add unemployment, employment, and related state economic controls.

Table A6. Placebo outcomes and multiple-testing correction

Source files:

Embedded table: Placebo Outcome Summary

outcome	kind	coef	se	t	n
share_any_exercise	placebo	-0.001	0.02	-0.051	251
share_flu_shot	placebo	-0.005	0.026	-0.195	251
share_fair_poor_health	primary	-0.02	0.012	-1.701	251
share_any_coverage	primary	0.009	0.015	0.558	251
share_medcost_any	primary	0.001	0.015	0.077	251

Embedded table: Romano Wolf Paper9

outcome	coef	se	t	n	p_conventional	p_romano_wolf	significant_conventional	significant_romano_wo
share_fair_poor_health	-0.02	0.012	-1.701	251	0.095	0.518	False	False
mean_mental_days	-0.698	0.476	-1.466	251	0.149	0.553	False	False
mean_phys_days	-0.24	0.361	-0.665	251	0.509	0.869	False	False
share_any_coverage	0.009	0.015	0.558	251	0.579	0.869	False	False
share_medcost_any	0.001	0.015	0.077	251	0.939	0.925	False	False

The placebo-outcome summary provides outcomes the unwinding should not move mechanically; the Romano-Wolf file reports family-wise adjusted p-values across the main BRFSS outcome family.

Table A7. HonestDiD sensitivity and Python-vs-R comparison

Source files:

Embedded table: Honestdid Acs Summary

sample	outcome	year	M	coef	se	max_pre_abs	pointwise_lower	pointwise_upper	honest_lower	honest_upper	width_ratio	zero_in_pointwise	zero_in_honest_ci
all_adults	has_medicaid	2023	0.5	0.003	0.006	0.018	-0.009	0.015	-0.019	0.024	4.23	True	True
all_adults	has_medicaid	2024	0.5	0.013	0.022	0.018	-0.031	0.057	-0.04	0.066	10.581	True	True
all_adults	has_medicaid	2023	1	0.003	0.006	0.018	-0.009	0.015	-0.028	0.033	6.08	True	True
all_adults	has_medicaid	2024	1	0.013	0.022	0.018	-0.031	0.057	-0.049	0.075	12.43	True	True
all_adults	has_medicaid	2023	2	0.003	0.006	0.018	-0.009	0.015	-0.046	0.052	9.779	True	True
all_adults	has_medicaid	2024	2	0.013	0.022	0.018	-0.031	0.057	-0.068	0.094	16.13	True	True
all_adults	uninsured	2023	0.5	-0.001	0.006	0.012	-0.012	0.01	-0.018	0.016	3.452	True	True
all_adults	uninsured	2024	0.5	-0.01	0.011	0.012	-0.032	0.012	-0.038	0.018	5.662	True	True
all_adults	uninsured	2023	1	-0.001	0.006	0.012	-0.012	0.01	-0.024	0.023	4.687	True	True
all_adults	uninsured	2024	1	-0.01	0.011	0.012	-0.032	0.012	-0.044	0.025	6.897	True	True
all_adults	uninsured	2023	2	-0.001	0.006	0.012	-0.012	0.01	-0.037	0.035	7.156	True	True
all_adults	uninsured	2024	2	-0.01	0.011	0.012	-0.032	0.012	-0.057	0.037	9.366	True	True
all_adults	has_any_coverage	2023	0.5	0.001	0.006	0.012	-0.01	0.012	-0.016	0.018	3.452	True	True
all_adults	has_any_coverage	2024	0.5	0.01	0.011	0.012	-0.012	0.032	-0.018	0.038	5.662	True	True
all_adults	has_any_coverage	2023	1	0.001	0.006	0.012	-0.01	0.012	-0.023	0.024	4.687	True	True
all_adults	has_any_coverage	2024	1	0.01	0.011	0.012	-0.012	0.032	-0.025	0.044	6.897	True	True
all_adults	has_any_coverage	2023	2	0.001	0.006	0.012	-0.01	0.012	-0.035	0.037	7.156	True	True
all_adults	has_any_coverage	2024	2	0.01	0.011	0.012	-0.012	0.032	-0.037	0.057	9.366	True	True
below_138fpl	has_medicaid	2023	0.5	0.041	0.027	0.019	-0.012	0.094	-0.021	0.104	12.493	True	True
below_138fpl	has_medicaid	2024	0.5	0.032	0.056	0.019	-0.077	0.142	-0.086	0.151	23.685	True	True
below_138fpl	has_medicaid	2023	1	0.041	0.027	0.019	-0.012	0.094	-0.031	0.113	14.361	True	True
below_138fpl	has_medicaid	2024	1	0.032	0.056	0.019	-0.077	0.142	-0.095	0.16	25.553	True	True
below_138fpl	has_medicaid	2023	2	0.041	0.027	0.019	-0.012	0.094	-0.049	0.132	18.097	True	True

sample	outcome	year	M	coef	se	max_pre_abs	pointwise_lower	pointwise_upper	honest_lower	honest_upper	width_ratio	zero_in_pointwise_ci	zero_in_honest_ci
below_138fpl	has_medicaid	2024	2	0.032	0.056	0.019	-0.077	0.142	-0.114	0.179	29.289	True	True
below_138fpl	uninsured	2023	0.5	-0.016	0.013	0.031	-0.041	0.008	-0.056	0.024	7.996	True	True
below_138fpl	uninsured	2024	0.5	-0.022	0.03	0.031	-0.081	0.037	-0.096	0.052	14.821	True	True
below_138fpl	uninsured	2023	1	-0.016	0.013	0.031	-0.041	0.008	-0.072	0.039	11.08	True	True
below_138fpl	uninsured	2024	1	-0.022	0.03	0.031	-0.081	0.037	-0.112	0.067	17.905	True	True
below_138fpl	uninsured	2023	2	-0.016	0.013	0.031	-0.041	0.008	-0.102	0.07	17.248	True	True
below_138fpl	uninsured	2024	2	-0.022	0.03	0.031	-0.081	0.037	-0.142	0.098	24.073	True	True
below_138fpl	has_any_coverage	2023	0.5	0.016	0.013	0.031	-0.008	0.041	-0.024	0.056	7.996	True	True
below_138fpl	has_any_coverage	2024	0.5	0.022	0.03	0.031	-0.037	0.081	-0.052	0.096	14.821	True	True
below_138fpl	has_any_coverage	2023	1	0.016	0.013	0.031	-0.008	0.041	-0.039	0.072	11.08	True	True
below_138fpl	has_any_coverage	2024	1	0.022	0.03	0.031	-0.037	0.081	-0.067	0.112	17.905	True	True
below_138fpl	has_any_coverage	2023	2	0.016	0.013	0.031	-0.008	0.041	-0.07	0.102	17.248	True	True
below_138fpl	has_any_coverage	2024	2	0.022	0.03	0.031	-0.037	0.081	-0.098	0.142	24.073	True	True

Embedded table: R Honestdid Acs Results

sample	outcome	year	coef	se	ci_lo_conv	ci_hi_conv	ci_lo_m05	ci_hi_m05	ci_lo_m1	ci_hi_m1	ci_lo_m2	ci_hi_m2	zero_in_honest_m
all_adults	has_medicaid	2023	0.003	0.006	-0.009	0.015	-0.018	0.023	-0.029	0.033	-0.055	0.059	TRUE
all_adults	has_medicaid	2024	0.013	0.022	-0.031	0.057	-0.048	0.066	-0.068	0.081	-0.118	0.124	TRUE
all_adults	uninsured	2023	-0.001	0.006	-0.012	0.01	-0.019	0.016	-0.029	0.024	-0.051	0.045	TRUE
all_adults	uninsured	2024	-0.01	0.011	-0.032	0.012	-0.047	0.02	-0.068	0.037	-0.112	0.079	TRUE
all_adults	has_any_coverage	2023	0.001	0.006	-0.01	0.012	-0.016	0.019	-0.024	0.029	-0.045	0.051	TRUE
all_adults	has_any_coverage	2024	0.01	0.011	-0.012	0.032	-0.02	0.047	-0.037	0.068	-0.079	0.112	TRUE
all_adults	has_private	2023	0.002	0.007	-0.013	0.016	-0.017	0.024	-0.027	0.035	-0.051	0.06	TRUE

sample	outcome	year	coef	se	ci_lo_conv	ci_hi_conv	ci_lo_m05	ci_hi_m05	ci_lo_m1	ci_hi_m1	ci_lo_m2	ci_hi_m2	zero_in_honest_m
all_adults	has_private	2024	0.003	0.016	-0.029	0.034	-0.035	0.051	-0.051	0.074	-0.098	0.125	TRUE
below_138fpl	has_medicaid	2023	0.041	0.027	-0.012	0.094	-0.028	0.113	-0.054	0.142	-0.124	0.213	TRUE
below_138fpl	has_medicaid	2024	0.032	0.056	-0.077	0.142	-0.113	0.168	-0.173	0.217	-0.315	0.355	TRUE
below_138fpl	uninsured	2023	-0.016	0.013	-0.041	0.008	-0.056	0.022	-0.08	0.045	-0.135	0.1	TRUE
below_138fpl	uninsured	2024	-0.022	0.03	-0.081	0.037	-0.107	0.059	-0.154	0.104	-0.262	0.21	TRUE
below_138fpl	has_any_coverage	2023	0.016	0.013	-0.008	0.041	-0.022	0.056	-0.045	0.08	-0.1	0.135	TRUE
below_138fpl	has_any_coverage	2024	0.022	0.03	-0.037	0.081	-0.059	0.107	-0.104	0.154	-0.21	0.262	TRUE
below_138fpl	has_private	2023	-0.018	0.02	-0.058	0.021	-0.077	0.039	-0.107	0.069	-0.177	0.139	TRUE
below_138fpl	has_private	2024	0.01	0.036	-0.06	0.081	-0.105	0.113	-0.17	0.171	-0.311	0.31	TRUE
below_138fpl	has_medicaid_expansion	2023	0.046	0.031	-0.015	0.108	-0.036	0.127	-0.07	0.16	-0.154	0.243	TRUE
below_138fpl	has_medicaid_expansion	2024	0.003	0.054	-0.104	0.11	-0.154	0.141	-0.23	0.204	-0.402	0.37	TRUE
below_138fpl	uninsured_expansion	2023	-0.017	0.015	-0.046	0.012	-0.058	0.021	-0.076	0.037	-0.119	0.079	TRUE
below_138fpl	uninsured_expansion	2024	-0.005	0.034	-0.072	0.062	-0.087	0.086	-0.119	0.12	-0.203	0.206	TRUE
below_138fpl	has_any_coverage_expansion	2023	0.017	0.015	-0.012	0.046	-0.021	0.058	-0.037	0.075	-0.079	0.119	TRUE
below_138fpl	has_any_coverage_expansion	2024	0.005	0.034	-0.062	0.072	-0.084	0.086	-0.12	0.119	-0.206	0.203	TRUE
below_138fpl	has_private_expansion	2023	-0.023	0.017	-0.057	0.011	-0.072	0.035	-0.097	0.07	-0.166	0.142	TRUE
below_138fpl	has_private_expansion	2024	0.019	0.027	-0.034	0.072	-0.069	0.119	-0.133	0.188	-0.275	0.332	TRUE

Embedded table: Python R Honestdid Acs Comparison

sample	outcome	year	coef_py	coef_r	coef_diff	se_py	se_r	se_diff	ci_lo_conv_py	ci_hi_conv_py	ci_lo_conv_r	ci_hi_conv_r	ci_lo_m05_py	ci_hi_m05_py	ci_lo_m05_r	ci_hi_m05_r	ci_lo_m05_diff	ci_hi_m05_diff	ci_lo_m1_py	ci_hi_m1_py	ci_lo_m1_r	ci_hi_m1_r	ci_lo_m1_diff	ci_hi_m1_diff	ci_lo_m2_py	ci_hi_m2_py	ci_lo_m2_r	ci_hi_m2_r	ci_lo_m2_diff	ci_hi_m2_diff	zero_in_m2_honest
all_adults	has_medicaid	2023	0.003	0.003		0.006	0.006	-0	-0.009	0.015	-0.009	0.015	-0.018	0.023	-0.019	0.024	0	-0.001	-0.029	0.033	-0.028	0.033	-0.002	0	-0.055	0.059	-0.046	0.052	-0.009	0.007	True
all_adults	has_medicaid	2024	0.013	0.013	-0	0.022	0.022		-0.031	0.057	-0.031	0.057	-0.048	0.066	-0.04	0.066	-0.008	0	-0.068	0.081	-0.049	0.075	-0.019	0.005	-0.118	0.124	-0.068	0.094	-0.051	0.03	True
all_adults	uninsured	2023	-0.001	-0.001		0.006	0.006		-0.012	0.01	-0.012	0.01	-0.019	0.016	-0.018	0.016	-0.001	-0.001	-0.029	0.024	-0.024	0.023	-0.005	0.001	-0.051	0.045	-0.037	0.035	-0.015	0.01	True

sample outcome year coef_diff se_diff … honest (column headers overprinted)														
all_uninsured adults 2024 - - 0.01 0.01	0.001 0.001	- 0.032	0.012 0.032	0.012 0.047	0.02 0.038	0.018 0.009	0.002 0.068	0.037 0.044	0.025 0.023	0.012 0.112	0.079 0.057	0.037 0.056	0.042	True
all_has_any_coverage 2023 0.001 0.001	0.006 0.006	- 0.01	0.012 0.01	0.012 0.016	0.019 0.016	0.018 0.001	0.001 0.024	0.029 0.023	0.024 0.001	0.005 0.045	0.051 0.035	0.037 0.01	0.015	True
all_has_any_coverage 2024 0.01 0.01	0.001 0.001	- 0.012	0.032 0.012	0.032 0.02	0.047 0.018	0.038 0.002	0.009 0.037	0.068 0.025	0.044 0.012	0.023 0.079	0.112 0.037	0.057 0.042	0.056	True
below_138fpl medicaid 2024 0.041 0	0.027 0.027 0	- 0.012	0.094 0.012	0.094 0.028	0.113 0.021	0.104 0.006	0.01 0.054	0.142 0.031	0.113 0.023	0.029 0.124	0.213 0.049	0.132 0.075	0.082	True
below_138fpl medicaid 2032 0	0.056 0.056	- 0.077	0.142 0.077	0.142 0.113	0.168 0.086	0.151 0.027	0.017 0.173	0.217 0.095	0.16 0.077	0.057 0.315	0.355 0.114	0.179 0.201	0.177	True
below_medicaid 2024 fpl 0 0.016 0.016	0.013 0.013	- 0.041	0.008 0.041	0.008 0.056	0.022 0.056	0.024 - 0.002	- 0.008	0.045 0.072	0.039 0.008	0.006 0.135	0.1- 0.102	0.07 0.032	0.03	True
below_medicaid 2024 fpl 0 0.022 0.022	0.03 0.03	- 0.081	0.037 0.081	0.037 0.107	0.059 0.096	0.052 0.011	0.007 0.154	0.104 0.112	0.067 0.042	0.036 0.262	0.21 0.142	0.098 0.12	0.112	True
below_138fpl_any_coverage 2016 0.016 0	0.013 0.013	- 0.008	0.041 0.008	0.041 0.022	0.056 0.024	0.056 0.002 0	- 0.045	0.08 0.039	0.072 0.006	0.008 0.1	0.135 0.07	0.102 0.03	0.032	True
below_138fpl_any_coverage 2022 0.022 0	0.03 0.03	- 0.037	0.081 0.037	0.081 0.059	0.107 0.052	0.096 0.007	0.011 0.104	0.154 0.067	0.112 0.036	0.042 0.21	0.262 0.098	0.142 0.112	0.12	True
below_138fpl medicaid_expansion 2016 0.031		- 0.015	0.108	- 0.036	0.127		- 0.07	0.16		- 0.154	0.243			True
below_138fpl medicaid_expansion 2013 0.054		- 0.104	0.11	- 0.154	0.141		- 0.23	0.204		- 0.402	0.37			True
below_medicaid 2024 fpl_expansion 0.015 0.017		- 0.046	0.012	- 0.058	0.021		- 0.076	0.037		- 0.119	0.079			True
below_medicaid 2024 fpl_expansion 0.034 0.005		- 0.072	0.062	- 0.087	0.086		- 0.119	0.12		- 0.203	0.206			True
below_138fpl_any_coverage 2017 expansion 0.015		- 0.012	0.046	- 0.021	0.058		- 0.037	0.075		- 0.079	0.119			True
below_138fpl_any_coverage 2005 expansion 0.034		- 0.062	0.072	- 0.084	0.086		- 0.12	0.119		- 0.206	0.203			True

These files record the pre-trend-sensitivity bounds and the formal R-package replication used to validate the manual implementation.

Table A8. Minimum detectable effect analysis

Source file:

Embedded table: Mde And Power

sample	outcome	n	n_clusters	residual_sd		n_eff	var_inter			unit								target_pp
BRFSS	share_fair_poor_health	251	51	0.108	0.645	51	0.015	0.027	0.834	pp	4	TennCare +4 pp fair/poor	adequately powered	7.88	Rome proxy	adequately powered	7.2	adequately powered for this effect size
BRFSS	share_any_coverage	251	51	0.01	0.73	51	0.015	0.036	1.114	pp	5	TennCare +5 pp uninsurance	adequately powered	7.88	Rome 3.94% p/p Rx proxy (~8 pp)	adequately powered	9.15	adequately powered for this effect size
BRFSS	share_med_cost_any	251	51	0.008	0.781	51	0.015	0.028	0.849	pp	3	TennCare implied +3 pp cost-barrier	adequately powered	3.94	Rome 3.94% p/p Rx	adequately powered	5	adequately powered for this effect size
BRFSS	mean_ment_health_days	251	51	0.237	0.64	51	0.015	0.85	0.261	days	1.5	OHIE proxy 1.5 days	adequately powered	1	Rome Rx proxy 1 day	adequately powered	1.5	adequately powered for this effect size
BRFSS	mean_phys_health_days	251	51	0.176	0.508	51	0.015	0.63	0.193	days	1	OHIE-style 1 day	adequately powered	0.75	Rome Rx proxy 0.75 day	adequately powered	1	adequately powered for this effect size

sample outcome n_clusters	residual_sd n_eff	var_inte... intensity_per_unit...	p90...	target...	10...	...comparison...	...target_pp	
ACS_hal_addicaid 11054088	0.3610.0192658	0.0150.1384.215pp	5	Tennessee +5 pp uninsurance	Adequately powered 7.88	Rome 3.94%/q Rx proxy (~8 pp)	adequately powered - 9.15	adequately powered for this effect size
ACS_allinsured 11054088	0.3230.0163179	0.0150.1133.454pp	5	Tennessee +5 pp uninsurance	Adequately powered 7.88	Rome 3.94%/q Rx proxy (~8 pp)	adequately powered - 9.15	adequately powered for this effect size
ACS_has_any_coverage 11054088	0.3230.0163179	0.0150.1133.454pp	5	Tennessee +5 pp uninsurance	Adequately powered 7.88	Rome 3.94%/q Rx proxy (~8 pp)	adequately powered - 9.15	adequately powered for this effect size
ACS_has_low_medicaid 13438392	0.4780.058878	0.0150.3179.726pp	5	Tennessee +5 pp uninsurance	Underpowered 7.88	Rome 3.94%/q Rx proxy (~8 pp)	underpowered - 9.15	underpowered
ACS_uninsured 13438392	0.4040.0371383	0.0150.2146.553pp	5	Tennessee +5 pp uninsurance	Underpowered 7.88	Rome 3.94%/q Rx proxy (~8 pp)	adequately powered - 9.15	underpowered
ACS_has_low_any_coverage 13438392	0.4040.0371383	0.0150.2146.553pp	5	Tennessee +5 pp uninsurance	Underpowered 7.88	Rome 3.94%/q Rx proxy (~8 pp)	adequately powered - 9.15	underpowered

sampleoutcome n_clustersresid_sdn_effvar_intensityper_unit_p90_intensity_10_pp_detectable_pp_anchor_comparisonstarget_pp						
ACS_has_low_med_9017580pl_0.4930.0351100 0.0150.2928.955pp	5	Tenn +5 pp unin- sur- ance	Cardle 7.88 underpowered	Rome 3.94%/q Rx proxy (~8 pp)	under- powered 9.15	underpowered
ACS_uninsured_9017580pl_0.3690.0211812 0.0150.17 5.221pp	5	Tenn +5 pp unin- sur- ance	Cardle 7.88 underpowered	Rome 3.94%/q Rx proxy (~8 pp)	adequately pow- ered 9.15	underpowered
ACS_has_any_coverage_9017580pl_0.3690.0211812 0.0150.17 5.221pp	5	Tenn +5 pp unin- sur- ance	Cardle 7.88 underpowered	Rome 3.94%/q Rx proxy (~8 pp)	adequately pow- ered 9.15	underpowered

This table formalizes the distinction between a precise population-level null and an underpowered low-income coverage analysis.

Appendix Figures

Figure A1. BRFSS event-study, main outcomes

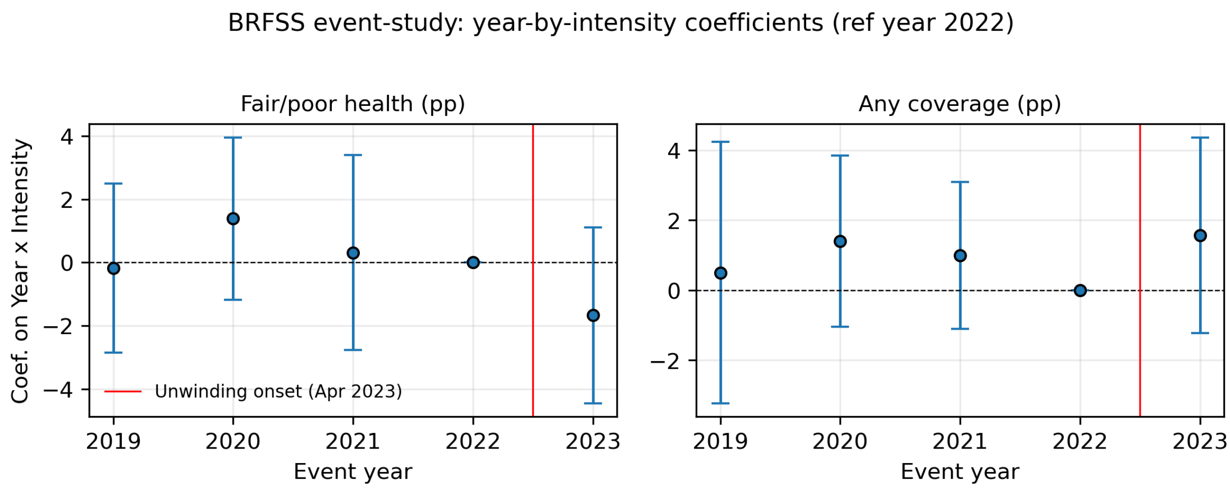

Figure 12: Brfss Event Study Main

This figure is the main BRFSS diagnostic for the continuous-intensity design and makes the limited pre/post structure transparent.

Figure A2. BRFSS event-study, share outcomes appendix panel

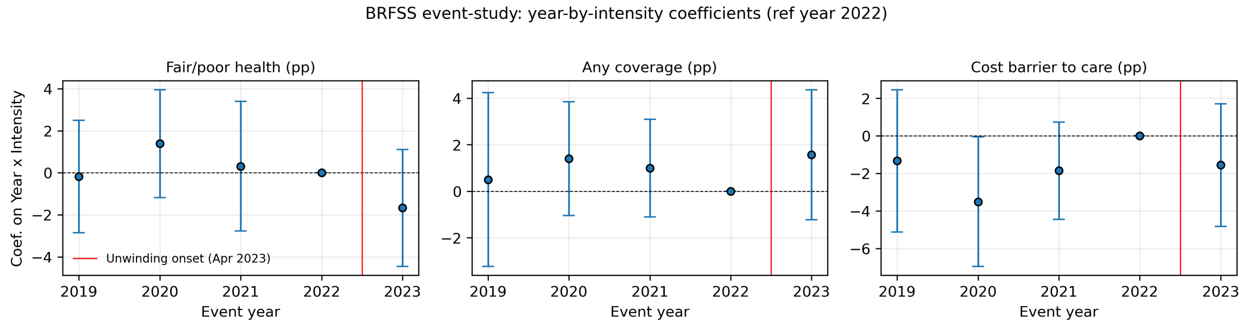

Figure 13: Brfss Event Study Shares App

Appendix panel for the share outcomes not carried in the main exhibit.

Figure A3. BRFSS event-study, unhealthy-days appendix panel

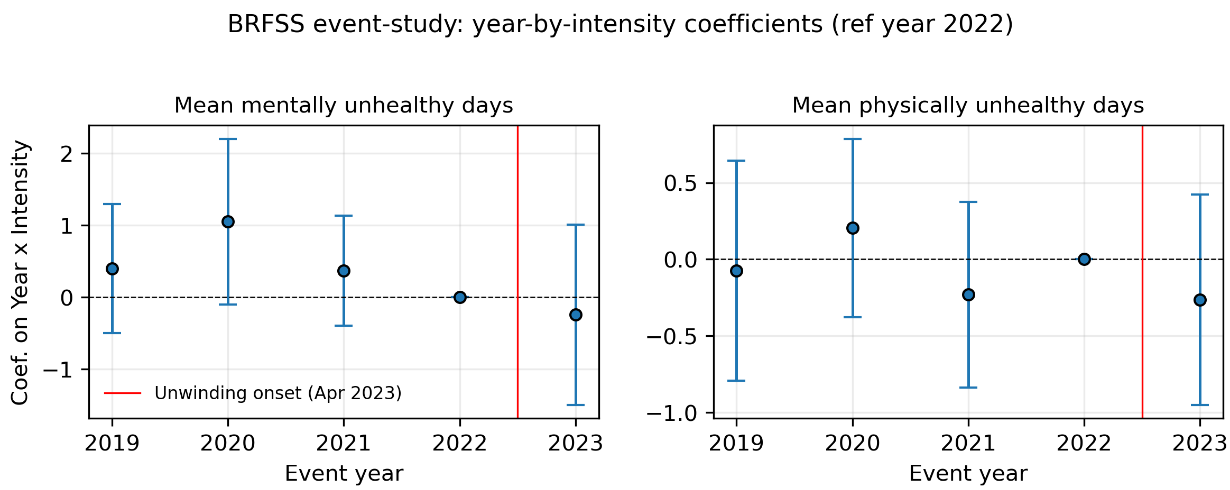

Figure 14: Brfss Event Study Days App

Appendix panel for the two unhealthy-days outcomes.

Figure A4. Treatment-intensity distribution

This is the dose-response analogue of a treatment-timing support plot in a staggered-adoption paper.

Figure A5. Tercile dose-response comparison

This figure checks whether the continuous-intensity null is masking a more threshold-like pattern across high-, medium-, and low-intensity states.

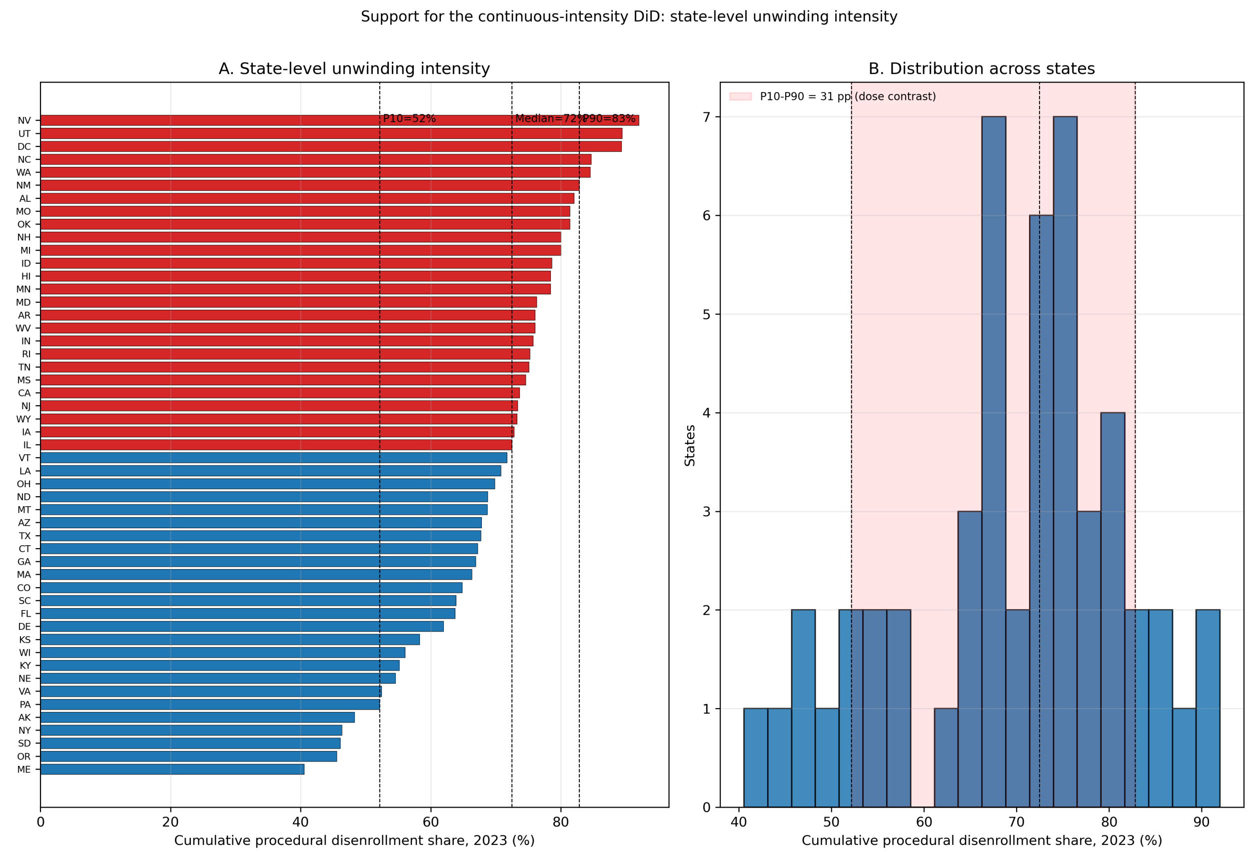

Figure 15: Intensity Distribution

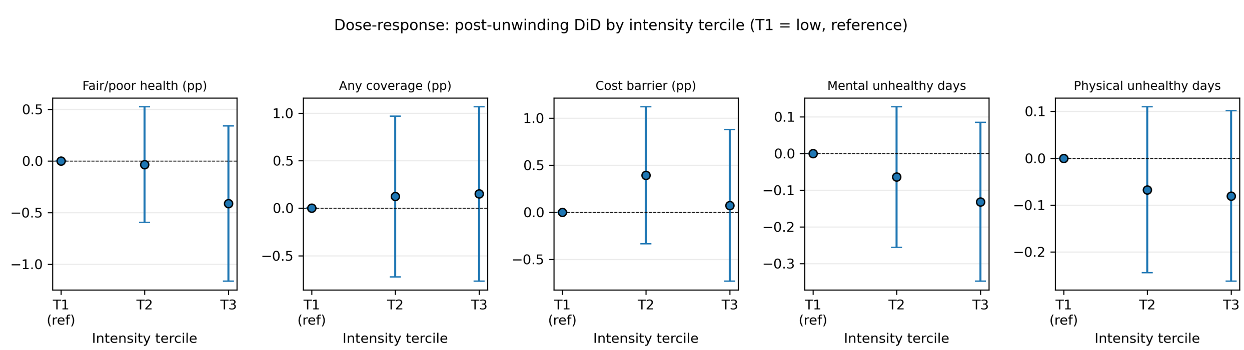

Figure 16: Dose Response Terciles

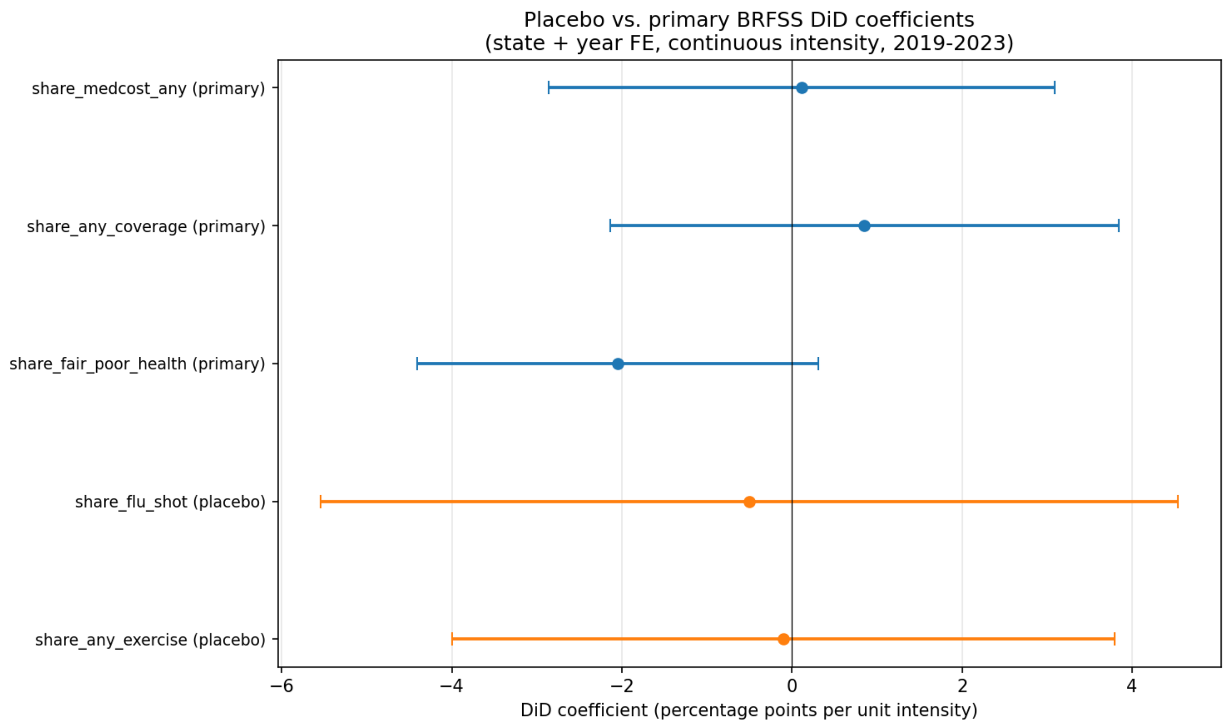

Figure 17: Placebo Outcome Comparison

Figure A6. Placebo outcomes versus primary outcomes

This figure packages the placebo outcomes directly against the main BRFSS estimates on a comparable scale.

Figure A7. ACS HonestDiD summary, low-income subsample

The low-income ACS panel is the paper's most policy-relevant coverage cross-check and the place where pre-trend-sensitivity bounds matter most.

Figure A8. ACS HonestDiD summary, all adults

This figure shows the same sensitivity logic for the all-adults ACS sample.

Figure A9. R HonestDiD replication panels

Representative panel from the full R replication set. The full grid remains in the replication workflow.

End of appendix. Main dissertation draft: the replication workflow.

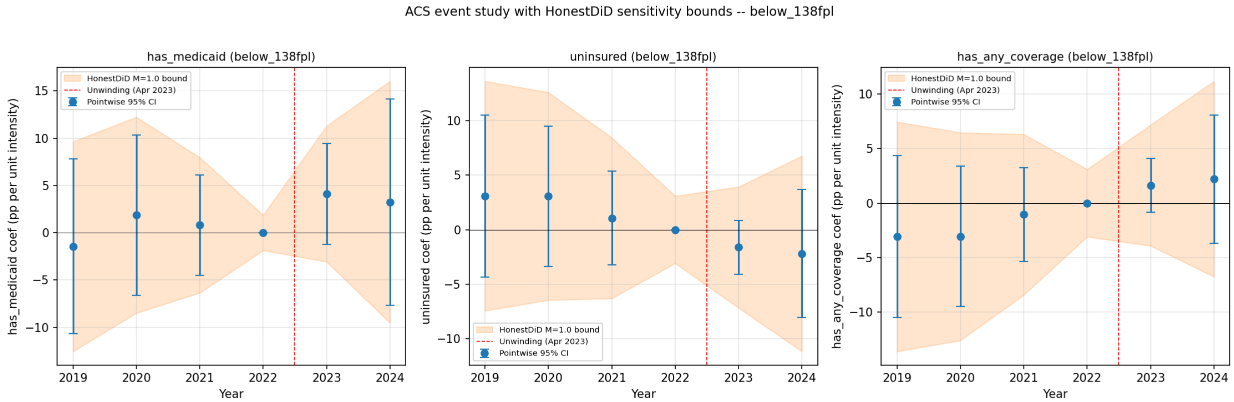

Figure 18: Acs Event Study Honestdid Below 138Fpl

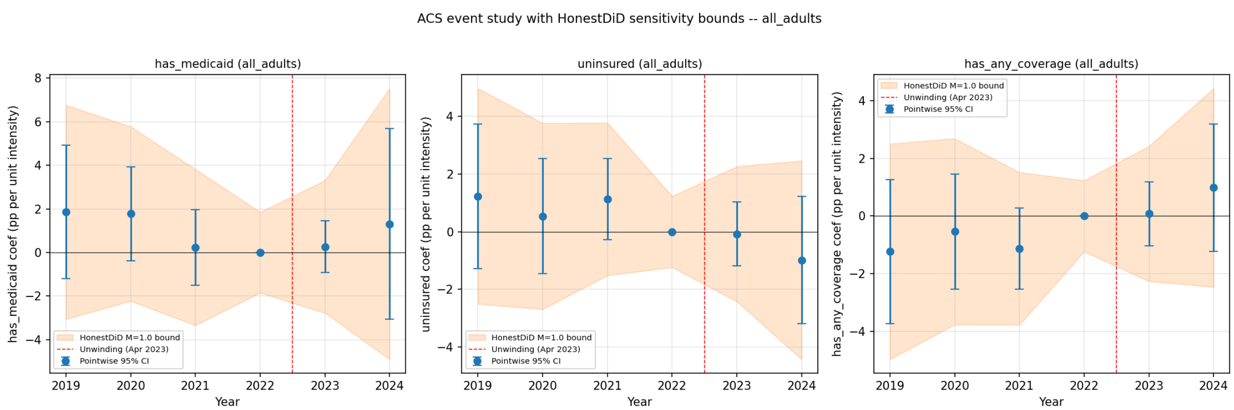

Figure 19: Acs Event Study Honestdid All Adults

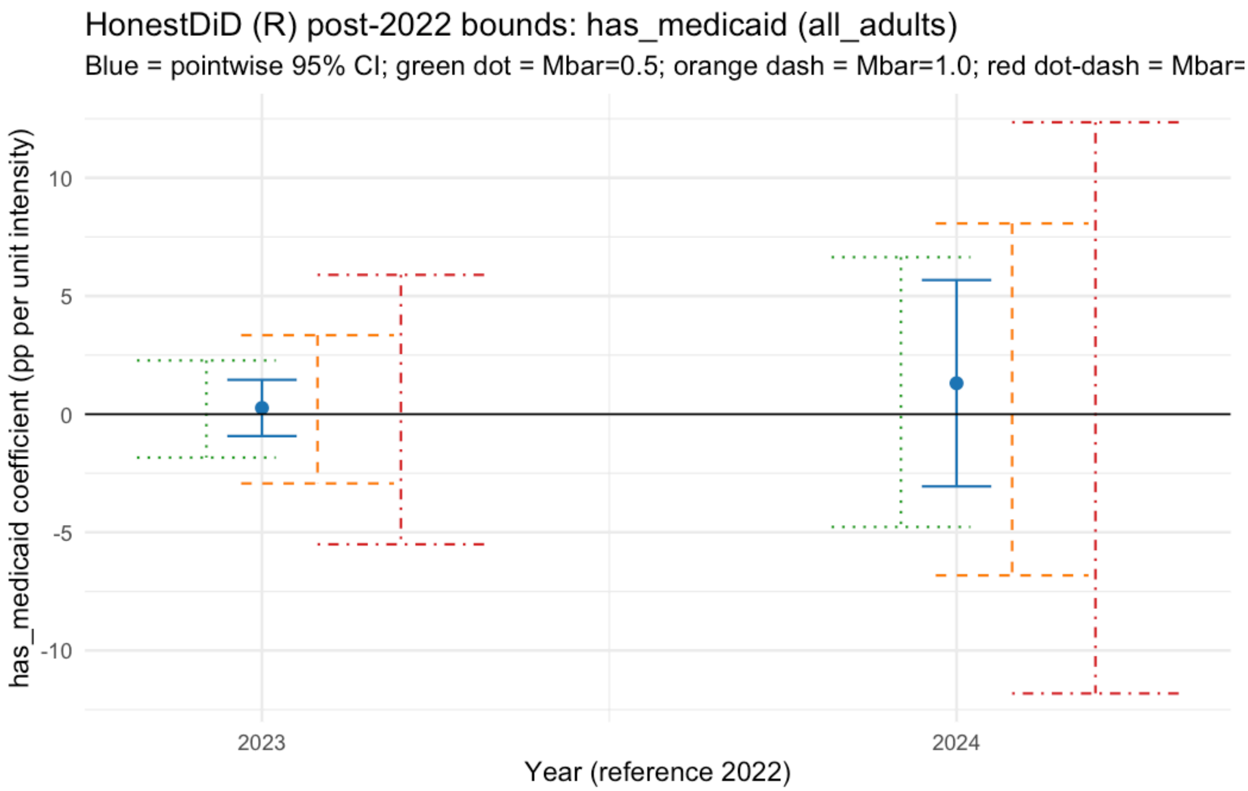

Figure 20: R Honestdid Has Medicaid All Adults